Supplementary Material

When Is Surrogate-Assisted Multiobjective Ensemble Weighting Trustworthy? A Replicated Study of Mixture Designs, Scheffé Surfaces and Normal Boundary Intersection on the Classifier-Weight Simplex

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Every item here is generated from the frozen study artifacts by `papers/surrogate_nbi_ensemble/build_supplement.py` and `build_assets.py`, at git tag `pco213-postwork-r30` (commit `b3ed050`) for the main study and the NSGA-II baseline commits that follow it. No number in this document is computed anywhere other than in the source CSVs named in each caption. Long tables are truncated where indicated; the complete table is always the named CSV, which is version-controlled.

Contents

S1 Extended protocol **2**

S2 Base models and single-objective references **3**

S3 Complete Scheffé results **5**

S4 All Pareto indicators **9**

S5 Full paired statistics **13**

S6 Reliability gate and regime analysis **20**

S7 NBI solver diagnostics **24**

S8 Empirical reference convergence **26**

S9 Cost definition and the accuracy–calibration conflict **27**

S10 Holdout transfer **30**

S11 From ten to thirty replications **33**

S12 Anchor-injection control **36**

S13 External NSGA-II baseline **37**

S14 Corrections log **43**

S1 Extended protocol

The design is a fixed 66-run mixture design over the five-component weight simplex: the union of the $\{5, 3\}$ and $\{5, 2\}$ simplex lattices, the overall centroid, the 5 axial points, the 5 quaternary centroids and the 10 midpoints between the overall centroid and the ternary centroids. It therefore contains the 5 pure vertices, all 10 binary edges, the ternary and quaternary faces, and uniform voting.

External validation uses 100 compositions per replication that are not in the design and never used for fitting: 60 from Dirichlet($\mathbf{1}$) (seed offset +1000) and 40 from Dirichlet($0.5 \cdot \mathbf{1}$) (offset +2000).

Every stochastic component is seeded as $20260904 + r$ for replication r: the outer stratified partition, the inner fold assignment, the model random states, the Dirichlet validation compositions (offsets +1000 and +2000), the direct AUC search (+3000, +4000), the empirical reference (+5000 onwards), the comparators (+12000), quality scoring (+13000) and NSGA-II (+770000). The seed policy, the full seed map, the configuration history and the resolved library versions are in `reports/pco213_postwork_benchmark/manifests/benchmark_manifest.json`; per-dataset provenance (source, acquisition command, the expected row, feature and positive-class-prevalence values, and the SHA-256 checksum of the file actually used) is in the corresponding `*_dataset_manifest.json`.

The statistical analysis plan is `STATISTICAL_ANALYSIS_R30.md`, first committed in `cf462b4` after replications 0–9 had been run and analysed, minutes after the extension was launched and before any replication of the extension had been completed; it is therefore prospective only for partitions 10–29. The tagged version (commit `b3ed050`) differs from it only in clarified wording and file references; no comparison, endpoint, test or family changed.

Base-model pipelines. Each base model is a `scikit-learn` pipeline whose preprocessing is fitted inside each training fold, and once on the whole training part for the holdout refit. Numeric features receive median imputation, with missing-value indicators on the two datasets that contain missing values (BNP Paribas and Porto Seguro), followed by standardization for logistic regression and k-nearest neighbours only. Categorical features keep a missing value as a level of its own and are one-hot encoded, with levels below 0.5% frequency grouped as infrequent, for logistic regression, Gaussian naive Bayes and k-nearest neighbours, or ordinal-encoded, with unseen levels mapped to -1, for the random forest and XGBoost; binary features are passed through. Logistic regression uses $C = 1$ with L-BFGS and at most 2,000 iterations, k-nearest neighbours brute-force Euclidean search with uniform weights, and XGBoost the histogram method without early stopping; all hyperparameters not listed here or in Section 4.2 of the main text are library defaults. The resolved values are listed under `model_params` in each dataset manifest, whose `random_state` of 42 is a placeholder: the runs seed the stochastic models with $20260904 + r$. Every component probability, out of fold and on the holdout, is clipped to $[10^{-3}, 1 - 10^{-3}]$ before any blend is formed.

Runner, checkpointing and execution provenance. The benchmark is driven by a single checkpointed runner (`scripts/pco213_run_postwork_benchmark.py`) with ten stages per replication: out-of-fold estimation, mixture design, Scheffé fitting, single-objective references, empirical Pareto reference, NBI-A, NBI-B, NBI-C, comparators and quality scoring. State is recorded per dataset × replication × stage, so an interrupted run resumes without recomputation and never silently recomputes a completed stage. A dry-run mode lists pending work; it was used to verify, before the extension was launched, that replications 0–9 were complete and that exactly 80 replications (800 stages) were pending. Aggregation and statistics are separate scripts (`pco213_postwork_benchmark_report.py` and `..._stats.py`), so the analysis can be rerun without re-executing the experiment. The library has 83 unit tests, including reference implementations checked against scikit-learn, exact hypervolume on known cases, the 66-run design, NBI anchor and warm-start behaviour, and the resumption planner, and the deterministic simplex repair and budget arithmetic of the NSGA-II baseline.

Execution consumed 81.1 hours of stage time (27.5 for the first ten replications, 53.6 for the extension) and produced 3,600 model fits, 19,920 design evaluations, 23,760 NBI subproblems, 5.02×

10^7 real objective evaluations, 1.55×10^7 reference points and 4.8×10^6 direct-AUC-search evaluations, on the environment recorded in the main text. Two stages were retried, both in replication 0 and none in the extension. The Porto Seguro comparator stage stopped because the simplex optimizer raised an error when no SLSQP start reported convergence at a vertex optimum; the optimizer was changed to compare its starts at their end points renormalized onto the simplex, to fall back to the best such point when no start reports convergence, and to evaluate the vertices and the centroid explicitly (commit `6fa7877`); the run was restarted, and one stage interrupted by the restart was rerun. The 32 stages of replication 0 completed before the change, under commit `c2e37c4`, were kept. On the stored surfaces the explicit vertex and centroid check changes none of their 8 surrogate optima and moves 64 of their 132 random-scalarization solutions onto the adjacent vertex by less than 10^{-10} in any weight. A moved solution can coincide with a point already on the reference front, so the replication-0 indicators could change slightly; they were not rescored with the moved solutions, and the effect of the other two changes on those stages cannot be checked without rerunning the optimizer. The extension ran under commit `5d1d5f4`, which changes only runner bookkeeping; the manifest's configuration history attributes the ten-replication configuration to that commit in error. The per-replication raw artifacts that are not version-controlled are the out-of-fold probability matrices, the candidate sets and the reference samples.

S2 Base models and single-objective references

Table S1: Base-model performance averaged over the 30 outer partitions of each dataset. Showing 7 of 16 columns; the rest are in the source CSV.

dataset	model	oof roc auc	holdout roc auc	oof log loss	oof brier	cost ms per 1k
bnp	gnb	0.603400	0.606100	3.741000	0.560200	3.696300
bnp	knn	0.677700	0.679900	0.512700	0.168900	149.632800
bnp	lr	0.734500	0.734700	0.484300	0.158500	3.319400
bnp	rf	0.742500	0.744600	0.482100	0.157300	6.191300
bnp	xgb	0.748300	0.753200	0.474000	0.154500	3.453500
porto	gnb	0.588400	0.592200	1.860500	0.320000	1.583300
porto	knn	0.566800	0.563600	0.161000	0.035200	230.531400
porto	lr	0.625100	0.627200	0.152900	0.034800	1.378500
porto	rf	0.620700	0.622400	0.153200	0.034900	4.343900
porto	xgb	0.591700	0.598500	0.158800	0.035300	1.641000
santander	gnb	0.888000	0.880400	0.211000	0.060500	0.784200
santander	knn	0.724700	0.727100	0.351500	0.092200	246.090200
santander	lr	0.859400	0.859300	0.232200	0.066600	0.720200
santander	rf	0.845100	0.845600	0.275200	0.079400	3.807000
santander	xgb	0.881100	0.883100	0.220100	0.063400	1.075700
uci_credit	gnb	0.658300	0.670900	1.160700	0.415000	0.567000
uci_credit	knn	0.757100	0.758400	0.445000	0.140400	24.117800
uci_credit	lr	0.723800	0.724500	0.465500	0.144900	0.515900
uci_credit	rf	0.782900	0.783200	0.427500	0.134200	5.666600
uci_credit	xgb	0.765100	0.766300	0.455300	0.140400	0.858900

Table S2: Single-objective reference solutions, averaged over 30 partitions. Showing 8 of 24 columns; the rest are in the source CSV.

dataset	method	oof roc auc	oof log loss	holdout roc auc	holdout log loss	cost weighted	cost support
bnp	best_single	0.748300	0.474000	0.753200	0.469700	3.453500	3.453500
bnp	cheapest_vertex	0.734500	0.484300	0.734700	0.484100	3.319400	3.319400
bnp	direct_auc_search	0.754700	0.469800	0.757300	0.468300	4.373900	13.089000
bnp	scheffe_optimum_auc	0.748800	0.474600	0.753300	0.471100	7.899400	35.169800

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dataset	method	oof roc auc	oof log loss	holdout roc auc	holdout log loss	cost weighted	cost support
bnp	scheffe_optimum_logloss	0.746100	0.483000	0.750700	0.479300	3.466900	4.566400
bnp	slsqp_direct_logloss	0.753900	0.468900	0.757000	0.466900	3.835300	16.660500
bnp	stacking_lr	0.754000	0.472000	0.756300	0.470500	–	–
bnp	uniform_voting	0.735800	0.518800	0.737200	0.517600	33.258700	166.293300
porto	best_single	0.625100	0.152900	0.627200	0.152700	1.378500	1.378500
porto	cheapest_vertex	0.625100	0.152900	0.627200	0.152700	1.378500	1.378500
porto	direct_auc_search	0.631100	0.152500	0.632500	0.152400	15.782700	238.000000
porto	scheffe_optimum_auc	0.629900	0.152500	0.631900	0.152400	2.687100	14.178500
porto	scheffe_optimum_logloss	0.618700	0.162200	0.620700	0.162000	4.112300	4.771600
porto	slsqp_direct_logloss	0.631100	0.152500	0.632500	0.152400	13.089700	238.105800
porto	stacking_lr	0.623900	0.153200	0.626100	0.153100	–	–
porto	uniform_voting	0.615100	0.196200	0.617900	0.195700	47.895600	239.478000
santander	best_single	0.888000	0.211000	0.880400	0.217500	0.784200	0.784200
santander	cheapest_vertex	0.859400	0.232200	0.859300	0.232200	0.720200	0.720200
santander	direct_auc_search	0.892300	0.212000	0.890000	0.213100	3.737100	185.244400
santander	scheffe_optimum_auc	0.890200	0.215500	0.885400	0.217500	45.755700	217.637600
santander	scheffe_optimum_logloss	0.890100	0.211100	0.884900	0.214100	26.578400	247.950000
santander	slsqp_direct_logloss	0.891700	0.208800	0.888000	0.211300	0.876000	1.859900
santander	stacking_lr	0.886800	0.215200	0.886300	0.216100	–	–
santander	uniform_voting	0.889300	0.226800	0.889200	0.226200	50.495500	252.477300
uci_credit	best_single	0.782900	0.427500	0.783200	0.427300	5.666600	5.666600
uci_credit	cheapest_vertex	0.723800	0.465500	0.724500	0.465200	0.515900	0.515900
uci_credit	direct_auc_search	0.783900	0.427300	0.784100	0.427100	6.867300	31.498500
uci_credit	scheffe_optimum_auc	0.783800	0.427300	0.784100	0.427100	7.671500	31.124900
uci_credit	scheffe_optimum_logloss	0.783100	0.427700	0.783300	0.427500	5.825900	8.610000
uci_credit	slsqp_direct_logloss	0.783600	0.426800	0.783700	0.426700	5.982700	31.196300
uci_credit	stacking_lr	0.782600	0.428600	0.782600	0.428600	–	–
uci_credit	uniform_voting	0.767100	0.465200	0.767800	0.465200	6.345300	31.726300

Table S3: Measured inference cost per base model and dataset, in milliseconds per 1,000 rows. First 60 of 600 rows; the complete table is the source CSV. Showing 3 of 4 columns; the rest are in the source CSV.

dataset	model	cost ms per 1k
santander	lr	0.716700
santander	gnb	0.778100
santander	knn	140.709600
santander	rf	3.802600
santander	xgb	1.091700
santander	lr	0.717000
santander	gnb	0.789200
santander	knn	254.039800
santander	rf	3.658800
santander	xgb	1.021200
santander	lr	0.717700
santander	gnb	0.781900
santander	knn	252.673600
santander	rf	3.793000
santander	xgb	1.039900
santander	lr	0.726900
santander	gnb	0.782400
santander	knn	251.787600
santander	rf	3.752300
santander	xgb	1.030300
santander	lr	0.715600
santander	gnb	0.775400
santander	knn	252.617800
santander	rf	3.589200
santander	xgb	1.040700
santander	lr	0.723900
santander	gnb	0.786500
santander	knn	247.097600
santander	rf	3.723300
santander	xgb	1.092700
santander	lr	0.714500
santander	gnb	0.770900
santander	knn	253.229800

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dataset	model	cost ms per 1k
santander	rf	3.577000
santander	xgb	1.078400
santander	lr	0.724800
santander	gnb	0.778200
santander	knn	250.165600
santander	rf	3.852300
santander	xgb	1.047400
santander	lr	0.716300
santander	gnb	0.785900
santander	knn	250.512300
santander	rf	3.852700
santander	xgb	1.043100
santander	lr	0.719700
santander	gnb	0.791300
santander	knn	250.027900
santander	rf	3.870900
santander	xgb	1.064300
santander	lr	0.730700
santander	gnb	0.787900
santander	knn	216.665700
santander	rf	4.516700
santander	xgb	1.068700
santander	lr	0.721200
santander	gnb	0.787700
santander	knn	250.295500
santander	rf	3.794000
santander	xgb	1.053200

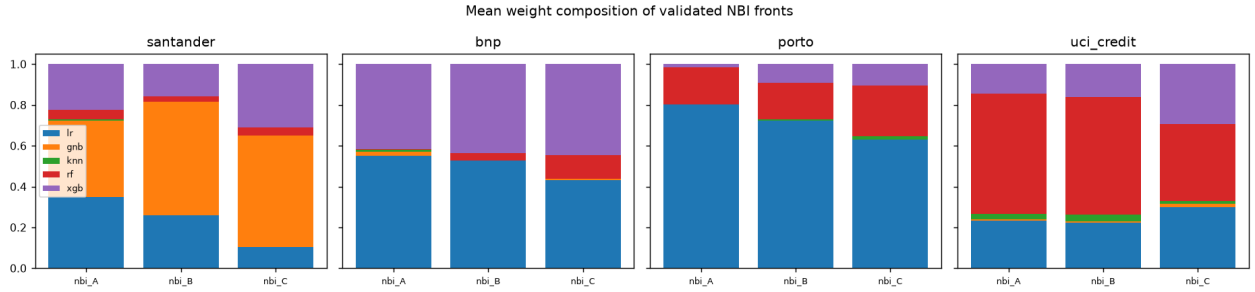


Figure S1: Weight composition of the solutions on the empirical reference front, by dataset.

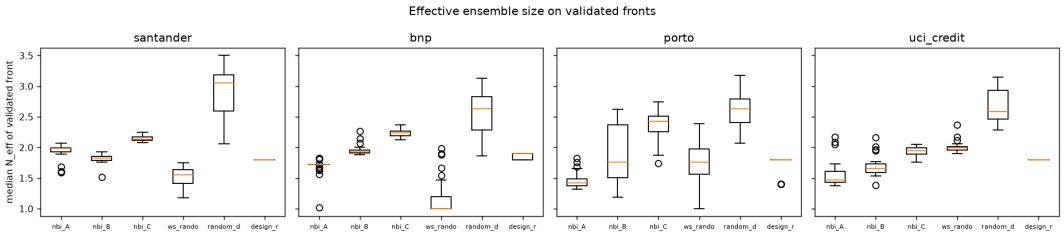


Figure S2: Effective number of models on the reference front.

S3 Complete Scheffé results

Table S4: Scheffé order comparison: how often each order is selected and passes the gate, with mean fit diagnostics over 30 partitions.

dataset	response	order	n	selected	reliable	r2 train	r2 external	rmse external	spearman	extrapolation
bnp	brier	linear	30	0	0	0.833300	0.836200	0.019600	0.965000	0.066200
bnp	brier	quadratic	30	30	30	1.000000	1.000000	0.000000	1.000000	0.000000
bnp	brier	special_cubic	30	0	0	1.000000	1.000000	0.000000	1.000000	0.000000
bnp	log_loss	linear	30	21	3	0.419400	-0.324100	0.128300	0.957500	0.069200
bnp	log_loss	quadratic	30	7	0	0.846400	-0.398600	0.141700	0.187300	0.000000
bnp	log_loss	special_cubic	30	2	0	0.893700	-0.331200	0.140300	0.615000	0.000000
bnp	pr_auc	linear	30	26	9	0.778200	0.482900	0.004200	0.932000	0.000000
bnp	pr_auc	quadratic	30	3	3	0.882000	0.408900	0.004500	0.933300	0.001300
bnp	pr_auc	special_cubic	30	1	1	0.913000	0.399000	0.004500	0.946700	0.000500
bnp	roc_auc	linear	30	22	1	0.673100	0.052500	0.009800	0.906200	0.001000
bnp	roc_auc	quadratic	30	7	3	0.874400	-0.022800	0.010100	0.894300	0.001000
bnp	roc_auc	special_cubic	30	1	1	0.901300	-0.033000	0.010100	0.915800	0.000300
porto	brier	linear	30	0	0	0.830800	0.834100	0.013800	0.995100	0.046700
porto	brier	quadratic	30	30	30	1.000000	1.000000	0.000000	1.000000	0.000000
porto	brier	special_cubic	30	0	0	1.000000	1.000000	0.000000	1.000000	0.000000
porto	log_loss	linear	30	20	12	0.535400	0.435700	0.061800	0.994600	0.038000
porto	log_loss	quadratic	30	8	0	0.884400	0.424200	0.065700	0.364900	0.000000
porto	log_loss	special_cubic	30	2	0	0.919700	0.446500	0.065100	0.787500	0.000000
porto	pr_auc	linear	30	3	2	0.773000	0.604300	0.001300	0.922800	0.000600
porto	pr_auc	quadratic	30	26	24	0.959000	0.766600	0.001000	0.934500	0.000300
porto	pr_auc	special_cubic	30	1	1	0.965800	0.776100	0.001000	0.937600	0.000300
porto	roc_auc	linear	30	2	0	0.731800	0.313900	0.004600	0.883300	0.002700
porto	roc_auc	quadratic	30	26	14	0.949200	0.529300	0.003800	0.921400	0.002000
porto	roc_auc	special_cubic	30	2	2	0.967000	0.572300	0.003700	0.940900	0.001500
santander	brier	linear	30	0	0	0.901900	0.925000	0.001400	0.985100	0.003300
santander	brier	quadratic	30	30	30	1.000000	1.000000	0.000000	1.000000	0.000000
santander	brier	special_cubic	30	0	0	1.000000	1.000000	0.000000	1.000000	0.000000
santander	log_loss	linear	30	0	0	0.797800	0.830100	0.006400	0.968800	0.012400
santander	log_loss	quadratic	30	29	29	0.986500	0.969900	0.002700	0.993400	0.000300
santander	log_loss	special_cubic	30	1	1	0.988800	0.970600	0.002700	0.994800	0.000100
santander	pr_auc	linear	30	4	0	0.432700	-0.908200	0.024800	0.598400	0.011200
santander	pr_auc	quadratic	30	26	0	0.878200	-0.339800	0.020500	0.768000	0.000000
santander	pr_auc	special_cubic	30	0	0	0.892200	-0.283900	0.020000	0.779700	0.000000
santander	roc_auc	linear	30	4	0	0.411000	-1.046300	0.010500	0.663700	0.006100
santander	roc_auc	quadratic	30	26	0	0.879600	-0.356500	0.008600	0.839400	0.000000
santander	roc_auc	special_cubic	30	0	0	0.897700	-0.343100	0.008600	0.859900	0.000000
uci_credit	brier	linear	30	0	0	0.827600	0.831200	0.013700	0.958600	0.046500
uci_credit	brier	quadratic	30	30	30	1.000000	1.000000	0.000000	1.000000	0.000000
uci_credit	brier	special_cubic	30	0	0	1.000000	1.000000	0.000000	1.000000	0.000000
uci_credit	log_loss	linear	30	0	0	0.810200	0.843100	0.030400	0.967000	0.092700
uci_credit	log_loss	quadratic	30	27	27	0.993800	0.991800	0.007000	0.990000	0.001200
uci_credit	log_loss	special_cubic	30	3	3	0.995800	0.992300	0.006900	0.996000	0.000600
uci_credit	pr_auc	linear	30	0	0	0.719300	0.709800	0.008900	0.916300	0.012300
uci_credit	pr_auc	quadratic	30	29	29	0.968900	0.916900	0.004600	0.981100	0.000800
uci_credit	pr_auc	special_cubic	30	1	1	0.972800	0.915400	0.004700	0.989000	0.000600
uci_credit	roc_auc	linear	30	0	0	0.861200	0.886400	0.005600	0.961600	0.007200
uci_credit	roc_auc	quadratic	30	20	20	0.995000	0.990100	0.001700	0.993700	0.000200
uci_credit	roc_auc	special_cubic	30	10	10	0.996200	0.991300	0.001500	0.995100	0.000200

Table S5: Full Scheffé coefficient stability: mean, dispersion, bootstrap interval, sign frequency, rank statistics, vertex gap, real 50/50 blend outcome and Pareto participation for every term. First 70 of 120 rows; the complete table is the source CSV. Showing 13 of 26 columns; the rest are in the source CSV.

dataset	response	term	mean	sd	cv	ci95 lo	ci95 hi	sign pos freq	top1 freq	vertex gap mean	blend50 minus best member mean	blend50 beats best member freq
santander	roc_auc	gnb	0.886300	0.000600	0.000600	0.886100	0.886500	1.000000	–	–	–	–
santander	roc_auc	gnb*knn	0.243200	0.002700	0.011100	0.242300	0.244200	1.000000	1.000000	0.163200	-0.005000	0.000000
santander	roc_auc	gnb*rf	0.075800	0.001200	0.015300	0.075400	0.076200	1.000000	0.000000	0.042900	0.000200	0.766700
santander	roc_auc	gnb*xgb	0.020800	0.000600	0.027200	0.020600	0.021000	1.000000	0.000000	0.006900	0.003900	1.000000
santander	roc_auc	knn	0.756500	0.001700	0.002300	0.755900	0.757200	1.000000	–	–	–	–
santander	roc_auc	knn*rf	0.199300	0.002100	0.010700	0.198500	0.200000	1.000000	0.000000	0.120300	0.007800	1.000000
santander	roc_auc	knn*xgb	0.232000	0.002600	0.011300	0.231100	0.232900	1.000000	0.000000	0.156400	-0.004100	0.000000
santander	roc_auc	lr	0.858000	0.000700	0.000900	0.857700	0.858300	1.000000	–	–	–	–
santander	roc_auc	lr*gnb	0.018200	0.000400	0.019600	0.018100	0.018400	1.000000	0.000000	0.028600	-0.007600	0.000000
santander	roc_auc	lr*knn	0.197200	0.002400	0.012000	0.196400	0.198000	1.000000	0.000000	0.134600	-0.002400	0.000000
santander	roc_auc	lr*rf	0.092700	0.001200	0.013500	0.092300	0.093200	1.000000	0.000000	0.014300	0.017800	1.000000
santander	roc_auc	lr*xgb	0.043400	0.000800	0.017700	0.043100	0.043700	1.000000	0.000000	0.021800	0.001100	1.000000
santander	roc_auc	rf	0.849300	0.001100	0.001200	0.848900	0.849700	1.000000	–	–	–	–
santander	roc_auc	rf*xgb	0.055500	0.001000	0.018200	0.055100	0.055800	1.000000	0.000000	0.036100	-0.001500	0.000000
santander	roc_auc	xgb	0.880300	0.000800	0.000900	0.880000	0.880600	1.000000	–	–	–	–
santander	log_loss	gnb	0.211400	0.000400	0.002100	0.211300	0.211600	1.000000	–	–	–	–
santander	log_loss	gnb*knn	-0.174000	0.001200	0.006800	-0.174400	-0.173600	0.000000	1.000000	0.140500	0.021000	0.000000
santander	log_loss	gnb*rf	-0.058000	0.000300	0.005500	-0.058100	-0.057900	0.000000	0.000000	0.064200	0.017400	0.000000
santander	log_loss	gnb*xgb	-0.020200	0.000200	0.010800	-0.020200	-0.020100	0.000000	0.000000	0.009100	-0.001500	1.000000
santander	log_loss	knn	0.340500	0.000700	0.002000	0.340300	0.340800	1.000000	–	–	–	–
santander	log_loss	knn*rf	-0.086600	0.001100	0.012700	-0.087000	-0.086200	0.000000	0.000000	0.076300	0.011100	0.000000
santander	log_loss	knn*xgb	-0.136700	0.001200	0.009000	-0.137200	-0.136300	0.000000	0.000000	0.131400	0.025500	0.000000
santander	log_loss	lr	0.232200	0.000400	0.001900	0.232100	0.232400	1.000000	–	–	–	–
santander	log_loss	lr*gnb	-0.013000	0.000100	0.010800	-0.013000	-0.012900	0.000000	0.000000	0.021200	0.006300	0.000000
santander	log_loss	lr*knn	-0.147600	0.001000	0.007100	-0.148000	-0.147200	0.000000	0.000000	0.119300	0.016700	0.000000
santander	log_loss	lr*rf	-0.053300	0.000300	0.005800	-0.053400	-0.053200	0.000000	0.000000	0.043000	0.007800	0.000000
santander	log_loss	lr*xgb	-0.029600	0.000300	0.009300	-0.029700	-0.029500	0.000000	0.000000	0.012100	-0.002300	1.000000
santander	log_loss	rf	0.275000	0.000300	0.001000	0.274900	0.275100	1.000000	–	–	–	–
santander	log_loss	rf*xgb	-0.041800	0.000400	0.008700	-0.042000	-0.041700	0.000000	0.000000	0.055100	0.016600	0.000000
santander	log_loss	xgb	0.220000	0.000600	0.002600	0.219800	0.220200	1.000000	–	–	–	–
bnp	roc_auc	gnb	0.637200	0.008200	0.012900	0.634200	0.640000	1.000000	–	–	–	–
bnp	roc_auc	gnb*knn	0.079800	0.016400	0.205800	0.074200	0.085800	1.000000	0.000000	0.074200	0.001000	0.900000
bnp	roc_auc	gnb*rf	0.138600	0.024100	0.174100	0.130400	0.147300	1.000000	0.000000	0.139000	-0.015600	0.000000
bnp	roc_auc	gnb*xgb	0.147800	0.025500	0.172500	0.139100	0.157000	1.000000	1.000000	0.144800	-0.018000	0.000000
bnp	roc_auc	knn	0.682000	0.001400	0.002000	0.681500	0.682500	1.000000	–	–	–	–
bnp	roc_auc	knn*rf	0.061600	0.001200	0.020000	0.061200	0.062000	1.000000	0.000000	0.064800	-0.012800	0.000000
bnp	roc_auc	knn*xgb	0.113100	0.002000	0.017800	0.112400	0.113800	1.000000	0.000000	0.070600	-0.004500	0.000000
bnp	roc_auc	lr	0.734300	0.000900	0.001200	0.734000	0.734600	1.000000	–	–	–	–
bnp	roc_auc	lr*gnb	0.132900	0.023700	0.178500	0.124900	0.141500	1.000000	0.000000	0.131100	-0.013700	0.000000
bnp	roc_auc	lr*knn	0.086000	0.001300	0.015600	0.085500	0.086400	1.000000	0.000000	0.056900	-0.003600	0.000000
bnp	roc_auc	lr*rf	0.010400	0.000700	0.069000	0.010200	0.010700	1.000000	0.000000	0.007900	0.002400	1.000000
bnp	roc_auc	lr*xgb	0.032400	0.001200	0.038300	0.032000	0.032900	1.000000	0.000000	0.013700	0.004100	1.000000
bnp	roc_auc	rf	0.743000	0.000900	0.001200	0.742600	0.743300	1.000000	–	–	–	–
bnp	roc_auc	rf*xgb	0.019100	0.001000	0.050200	0.018800	0.019500	1.000000	0.000000	0.005800	0.005300	1.000000
bnp	roc_auc	xgb	0.747100	0.001500	0.002000	0.746600	0.747700	1.000000	–	–	–	–
bnp	log_loss	gnb	3.035500	0.166800	0.055000	2.978200	3.095200	1.000000	–	–	–	–
bnp	log_loss	gnb*knn	-3.888100	0.242000	0.062200	-3.974600	-3.805000	0.000000	0.000000	3.228300	0.202500	0.000000
bnp	log_loss	gnb*rf	-3.891700	0.245800	0.063200	-3.979700	-3.807200	0.000000	0.000000	3.258900	0.217100	0.000000
bnp	log_loss	gnb*xgb	-3.921400	0.248400	0.063300	-4.010300	-3.836000	0.000000	1.000000	3.267000	0.214200	0.000000
bnp	log_loss	knn	0.471300	0.002800	0.005900	0.470300	0.472300	1.000000	–	–	–	–
bnp	log_loss	knn*rf	0.370700	0.025500	0.068700	0.362000	0.379800	1.000000	0.000000	0.030600	0.011300	0.000000

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dataset	response	term	mean	sd	cv	ci95 lo	ci95 hi	sign pos freq	top1 freq	vertex gap mean	blend50 minus best member mean	blend50 beats best member freq
bnp	log_loss	knn*xgb	0.332000	0.025700	0.077500	0.323200	0.341200	1.000000	0.000000	0.038700	0.005700	0.000000
bnp	log_loss	lr	0.442600	0.002700	0.006100	0.441600	0.443500	1.000000	–	–	–	–
bnp	log_loss	lr*gnb	-3.871600	0.248600	0.064200	-3.960600	-3.786100	0.000000	0.000000	3.256600	0.220900	0.000000
bnp	log_loss	lr*knn	0.353000	0.025600	0.072600	0.344200	0.362200	1.000000	0.000000	0.028300	0.005800	0.000000
bnp	log_loss	lr*rf	0.368500	0.025600	0.069400	0.359700	0.377600	1.000000	0.000000	0.002300	-0.003400	1.000000
bnp	log_loss	lr*xgb	0.351400	0.025600	0.072900	0.342700	0.360600	1.000000	0.000000	0.010400	-0.003600	1.000000
bnp	log_loss	rf	0.441000	0.002600	0.005900	0.440100	0.441900	1.000000	–	–	–	–
bnp	log_loss	rf*xgb	0.357400	0.025500	0.071500	0.348600	0.366600	1.000000	0.000000	0.008100	-0.003300	1.000000
bnp	log_loss	xgb	0.432200	0.002700	0.006200	0.431200	0.433100	1.000000	–	–	–	–
porto	roc_auc	gnb	0.593900	0.002200	0.003600	0.593100	0.594600	1.000000	–	–	–	–
porto	roc_auc	gnb*knn	0.070700	0.008100	0.114200	0.067800	0.073500	1.000000	0.633300	0.021700	0.009800	1.000000
porto	roc_auc	gnb*rf	0.008000	0.008800	1.089400	0.005100	0.011200	0.800000	0.000000	0.032200	-0.009500	0.000000
porto	roc_auc	gnb*xgb	0.047200	0.008600	0.181300	0.044300	0.050200	1.000000	0.000000	0.004100	0.013200	1.000000
porto	roc_auc	knn	0.568600	0.003100	0.005400	0.567600	0.569700	1.000000	–	–	–	–
porto	roc_auc	knn*rf	0.051700	0.002500	0.047700	0.050800	0.052500	1.000000	0.000000	0.053900	-0.012900	0.000000
porto	roc_auc	knn*xgb	0.067900	0.003400	0.050400	0.066700	0.069100	1.000000	0.366700	0.024900	0.005000	1.000000
porto	roc_auc	lr	0.625500	0.002600	0.004100	0.624500	0.626400	1.000000	–	–	–	–
porto	roc_auc	lr*gnb	0.001500	0.009000	6.200700	-0.001500	0.004700	0.466700	0.000000	0.036600	-0.012900	0.000000
porto	roc_auc	lr*knn	0.062800	0.002700	0.043400	0.061900	0.063800	1.000000	0.000000	0.058400	-0.012300	0.000000

The edge criterion. For a maximized response such as ROC-AUC, on the edge i–j the quadratic model gives $\hat{y}(t) = \beta_j + (\beta_i - \beta_j)t + \beta_{ij}t(1-t)$ with $t = w_i$, which has an interior maximum, at $t^* = (\beta_i - \beta_j + \beta_{ij})/(2\beta_{ij})$, if and only if $\beta_{ij} > |\beta_i - \beta_j|$; there $\hat{y}(t^*) - \max(\beta_i, \beta_j) = (\beta_{ij} - |\beta_i - \beta_j|)^2/(4\beta_{ij}) > 0$.

Coefficient stability and the blend test, per cell. The largest-magnitude interaction is the same term at $R = 10$ and $R = 30$ in all eight dataset–response cells and is the per-partition largest in 30 of 30 partitions in seven of them; on Porto Seguro's ROC-AUC surface two k-nearest-neighbour pairs alternate, 19 of 30 against 11 of 30. Between-partition coefficients of variation for the largest-magnitude term of each cell lie between 0.007 and 0.17. The largest interaction attaches to the weakest vertex of its pair: k-nearest neighbours on Santander and Porto Seguro, Gaussian naive Bayes on BNP Paribas and UCI credit. It tracks the vertex-quality gap of that pair (0.163, 0.145, 0.022 and 0.107). For that pair the real out-of-fold 50/50 blend minus its better member is −0.0050 (Santander), −0.0180 (BNP Paribas), +0.0098 (Porto Seguro) and −0.0201 (UCI credit) ROC-AUC, and the untouched holdout repeats the pattern (0/30, 0/30, 29/30, 0/30). Across all ten pairs of each dataset the criterion and the real 50/50 blend agree on 6, 4, 7 and 3 pairs; all 20 disagreements are the surface predicting a superior blend that the midpoint mixture does not deliver, a tally that tests only the midpoint of each edge and therefore overstates over-prediction.

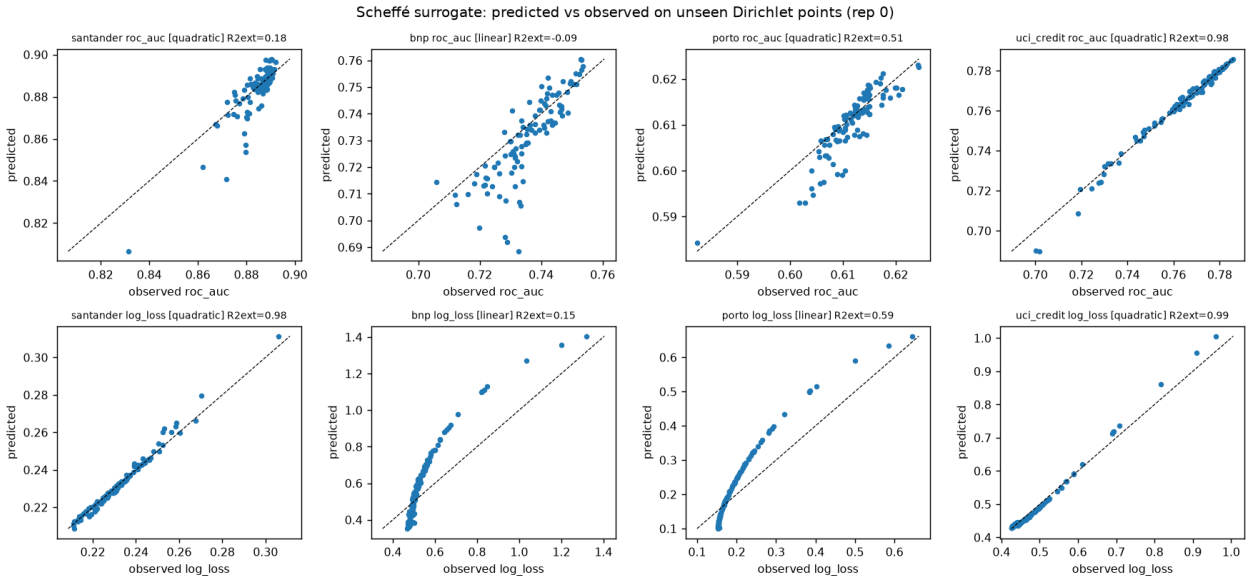


Figure S3: Predicted versus observed response on the 100 unseen compositions, all dataset–response pairs. Extends Figure 2 of the main text.

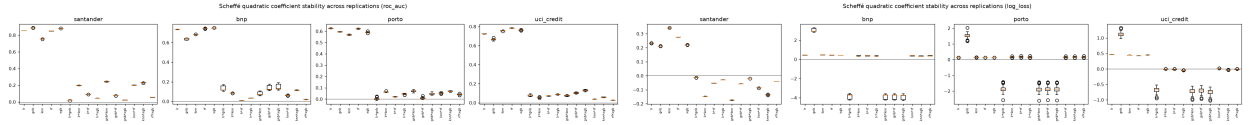


Figure S4: Coefficient stability across the 30 partitions, ROC-AUC (left) and log-loss (right).

S4 All Pareto indicators

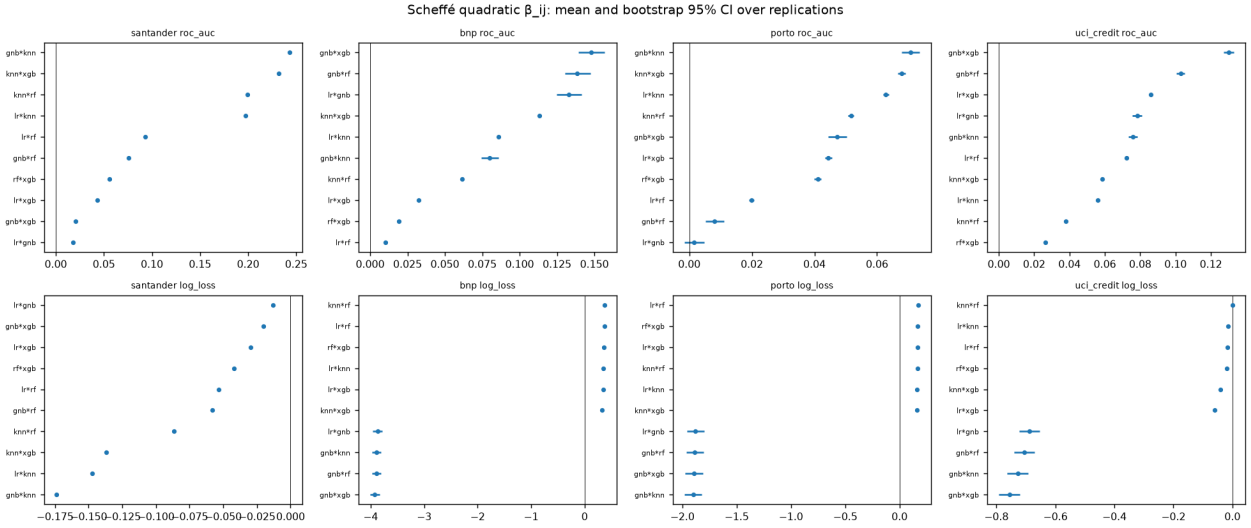


Figure S5: Interaction-coefficient bootstrap intervals across partitions.

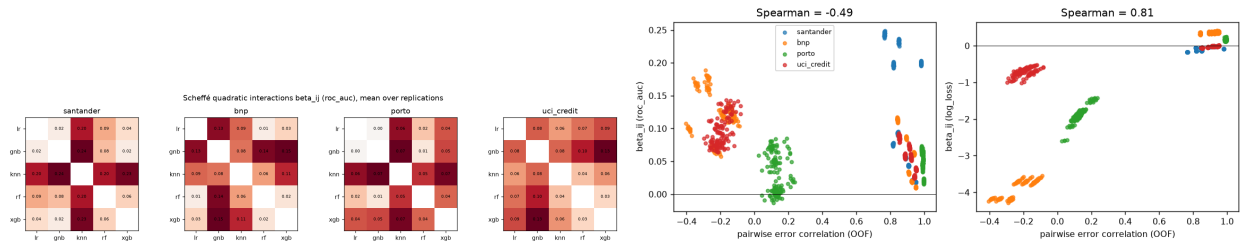


Figure S6: Left: interaction coefficients for ROC-AUC. Right: interaction magnitude against base-model error correlation, which the main text reads as a vertex-quality effect rather than complementarity.

Table S6: All Pareto quality indicators, median over 30 partitions, for every candidate set under both cost definitions. Showing 12 of 30 columns; the rest are in the source CSV.

cost	dataset	set	n front	gd front	igd	igd plus	spacing	spacing cv	hv ratio	joint nondominated fraction front	coverage set over ref
support	bnp	design_runs	5.000000	0.009600	0.079200	0.036300	0.531800	1.211500	0.935500	0.400000	0.000000
support	bnp	nbi_A	27.000000	0.004400	0.280300	0.040500	0.264600	5.196200	0.904300	0.043500	0.000000
support	bnp	nbi_B	10.000000	0.010700	0.059800	0.018600	0.473100	2.038800	0.970600	0.696200	0.000000
support	bnp	nbi_C	13.000000	0.006800	0.071900	0.029800	0.506700	3.021600	0.968600	0.769200	0.000000
support	bnp	random_dirichlet_budget	2.000000	2.514800	1.004700	0.983200	0.803800	1.140000	0.050400	0.000000	0.000000
support	bnp	single_objective_refs	6.000000	0.001100	0.066600	0.027000	0.542300	1.524300	0.981100	0.333300	0.000000
support	bnp	ws_random_scalarization	66.000000	0.005700	0.428800	0.349600	0.000000	0.738900	0.523800	0.000000	0.000000
support	porto	design_runs	3.000000	0.070500	0.567300	0.148500	0.599600	0.768500	0.709600	0.000000	0.000000
support	porto	nbi_A	3.000000	0.045900	0.540700	0.089400	0.540700	0.953000	0.839400	0.000000	0.000000
support	porto	nbi_B	6.000000	0.027500	0.058000	0.016600	0.455400	1.300300	0.869000	0.225000	0.000000
support	porto	nbi_C	12.500000	0.010800	0.074200	0.006900	0.337100	2.058800	0.933400	0.769200	0.000000
support	porto	random_dirichlet_budget	2.000000	6.847800	1.247400	1.232500	47.511500	1.538700	0.000000	0.000000	0.000000
support	porto	single_objective_refs	5.000000	0.005700	0.099900	0.012900	0.528900	2.163600	0.859500	0.400000	0.000000
support	porto	ws_random_scalarization	4.500000	0.047000	0.746700	0.091600	0.431900	1.999300	0.830700	0.000000	0.000000
support	santander	design_runs	4.000000	0.003000	0.337500	0.008000	0.847300	1.666100	0.987900	0.291700	0.000000
support	santander	nbi_A	8.000000	0.079200	0.071500	0.067200	0.444700	1.838200	0.853100	0.000000	0.000000
support	santander	nbi_B	11.000000	0.020000	0.024800	0.008200	0.440800	2.011200	0.986200	0.117600	0.000000
support	santander	nbi_C	29.500000	0.018000	0.013600	0.001500	0.256500	3.166900	0.995000	0.862100	0.000000
support	santander	random_dirichlet_budget	4.000000	0.245900	0.680300	0.662900	0.358700	1.072700	0.077400	0.000000	0.000000
support	santander	single_objective_refs	4.000000	0.001000	0.050500	0.011300	0.779500	0.930600	0.984200	0.500000	0.000000
support	santander	ws_random_scalarization	11.500000	0.003800	0.351300	0.009400	0.016000	1.802400	0.986200	0.000000	0.000000
support	uci_credit	design_runs	4.000000	0.002300	0.314700	0.012400	0.221400	0.338700	0.975000	0.450000	0.000000
support	uci_credit	nbi_A	8.500000	0.027500	0.097500	0.040000	0.395500	2.048800	0.910700	0.154800	0.000000
support	uci_credit	nbi_B	14.000000	0.024100	0.099800	0.049000	0.290000	2.262600	0.895400	0.455600	0.000000
support	uci_credit	nbi_C	16.500000	0.008000	0.044300	0.036600	0.373700	2.770100	0.941600	0.666700	0.000000
support	uci_credit	random_dirichlet_budget	2.000000	0.131200	0.567300	0.552700	1.149100	1.501800	0.097300	0.000000	0.000000
support	uci_credit	single_objective_refs	6.000000	0.001500	0.114000	0.037000	0.866300	2.285800	0.905400	0.333300	0.000000
support	uci_credit	ws_random_scalarization	15.000000	0.001300	0.149200	0.005600	0.096700	2.869500	0.983100	0.519200	0.000000
weighted	bnp	design_runs	6.000000	0.081900	0.169800	0.028900	0.522500	1.050300	0.918100	0.333300	0.000000
weighted	bnp	nbi_A	8.000000	0.001600	0.189200	0.027100	0.207600	0.827200	0.913000	0.625000	0.000000
weighted	bnp	nbi_B	44.000000	0.000400	0.055700	0.012000	0.126700	2.947700	0.971000	0.839100	0.000000
weighted	bnp	nbi_C	34.000000	0.008500	0.027500	0.013600	0.077600	0.889400	0.982900	0.558800	0.000000
weighted	bnp	random_dirichlet_budget	5.000000	5.072500	1.551000	1.525400	3.632400	0.989600	0.000000	0.000000	0.000000
weighted	bnp	single_objective_refs	4.000000	0.023100	0.204700	0.163400	0.442100	0.500100	0.759100	0.500000	0.000000
weighted	bnp	ws_random_scalarization	1.000000	0.086000	0.457800	0.307800	0.036200	1.575400	0.473700	0.000000	0.000000
weighted	porto	design_runs	3.000000	0.070700	0.354200	0.153800	0.594100	0.776600	0.647200	0.000000	0.000000
weighted	porto	nbi_A	11.000000	0.036800	0.250100	0.097400	0.067200	0.502400	0.769000	0.035400	0.000000
weighted	porto	nbi_B	59.500000	0.015500	0.059900	0.035400	0.063200	2.570200	0.913700	0.486800	0.000000
weighted	porto	nbi_C	56.000000	0.003300	0.027000	0.007200	0.047300	1.558300	0.982000	0.831900	0.000000
weighted	porto	random_dirichlet_budget	5.000000	47.584300	1.821600	1.792400	32.796800	1.125900	0.000000	0.000000	0.000000
weighted	porto	single_objective_refs	5.000000	0.009100	0.233900	0.082700	0.395400	1.440100	0.817500	0.400000	0.000000
weighted	porto	ws_random_scalarization	12.500000	0.036200	0.329600	0.104300	0.110800	2.138900	0.761800	0.000000	0.000000
weighted	santander	design_runs	7.000000	0.001900	0.090000	0.011300	0.309000	1.002500	0.980500	0.571400	0.000000
weighted	santander	nbi_A	13.000000	0.614900	0.148700	0.136200	0.991000	1.694700	0.788700	0.000000	0.000000
weighted	santander	nbi_B	19.500000	0.011300	0.042000	0.010500	0.148600	1.036100	0.980700	0.155000	0.000000
weighted	santander	nbi_C	35.000000	0.006800	0.043000	0.009500	0.198600	2.377200	0.989000	0.682500	0.000000
weighted	santander	random_dirichlet_budget	6.000000	3.826000	0.767400	0.714800	1.525000	0.934900	0.214600	0.000000	0.000000
weighted	santander	single_objective_refs	4.000000	0.000000	0.126400	0.014600	0.747300	0.893200	0.971500	0.750000	0.000000
weighted	santander	ws_random_scalarization	57.000000	0.000900	0.184900	0.007300	0.015400	2.845700	0.978200	0.868500	0.000000
weighted	uci_credit	design_runs	8.000000	0.012600	0.082800	0.032400	0.300900	1.087300	0.921600	0.333300	0.000000
weighted	uci_credit	nbi_A	11.000000	0.007000	0.249200	0.216500	0.121300	1.393600	0.621500	0.475000	0.000000

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cost	dataset	set	n front	gd front	igd	igd plus	spacing	spacing cv	hv ratio	joint nondominated fraction front	coverage set over ref
weighted	uci_credit	nbi_B	27.000000	0.002300	0.180400	0.154300	0.061500	1.948900	0.706200	0.561100	0.000000
weighted	uci_credit	nbi_C	65.000000	0.001400	0.028700	0.009500	0.031200	1.919100	0.975800	0.826500	0.000000
weighted	uci_credit	random_dirichlet_budget	9.500000	0.293000	0.238100	0.149600	0.380800	0.939900	0.733200	0.000000	0.000000
weighted	uci_credit	single_objective_refs	5.000000	0.023700	0.430500	0.402200	1.136000	1.824800	0.336800	0.366700	0.000000
weighted	uci_credit	ws_random_scalarization	43.500000	0.000500	0.135800	0.008600	0.029200	1.510600	0.978200	0.840000	0.000000

Per-partition counts behind the median levels. NBI-C's weighted hypervolume ratio exceeds 0.97 in every partition of Santander and BNP Paribas but in only 24 of 30 partitions of Porto Seguro and 25 of 30 of UCI credit; it exceeds 0.95 in all thirty partitions of Santander, BNP Paribas and UCI credit and in 26 of 30 on Porto Seguro, the worst partition being 0.877. The 66-point random Dirichlet comparator has a hypervolume ratio of exactly zero in 10 of 30 partitions on Santander, 21 of 30 on BNP Paribas and 25 of 30 on Porto Seguro.

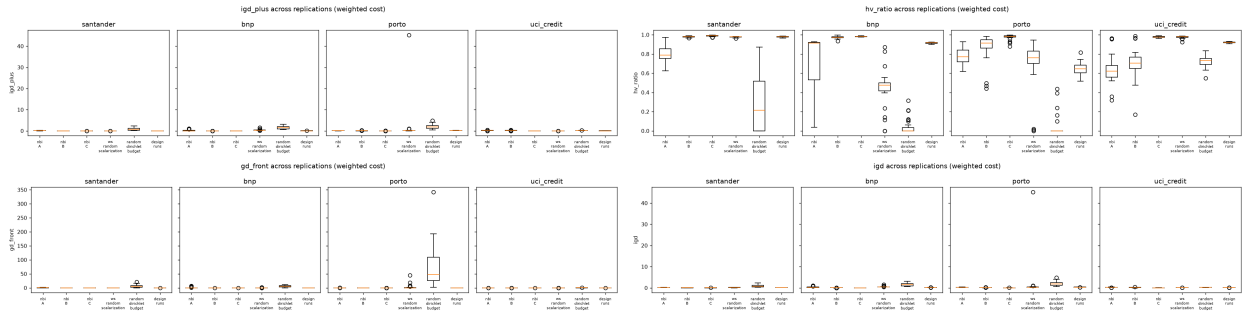


Figure S7: Distributions over the 120 replications of IGD^+, hypervolume ratio, generational distance and IGD.

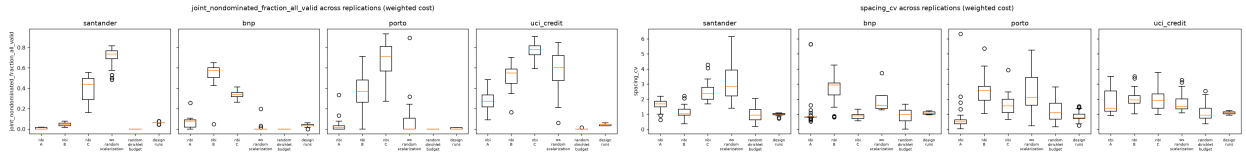


Figure S8: Joint non-dominated fraction and spacing coefficient of variation.

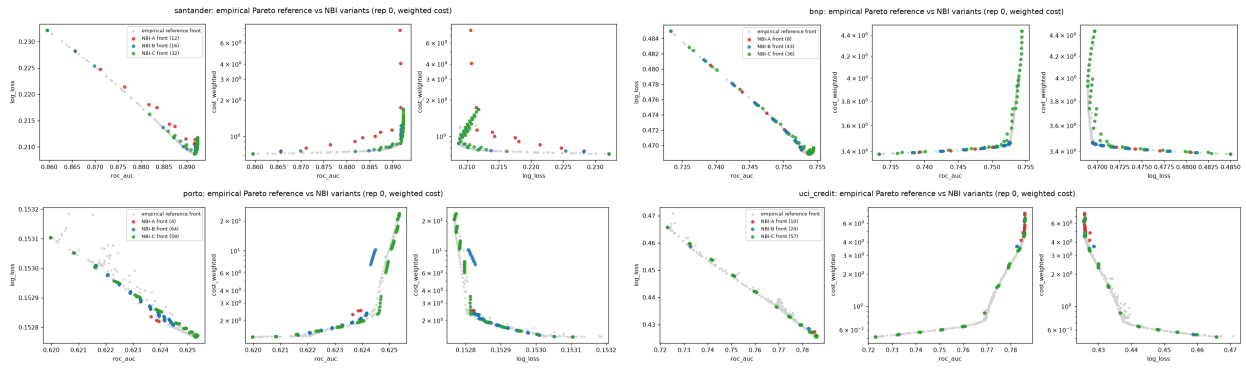


Figure S9: Revalidated fronts against the empirical reference, all four datasets. Extends Figure 3 of the main text.

S5 Full paired statistics

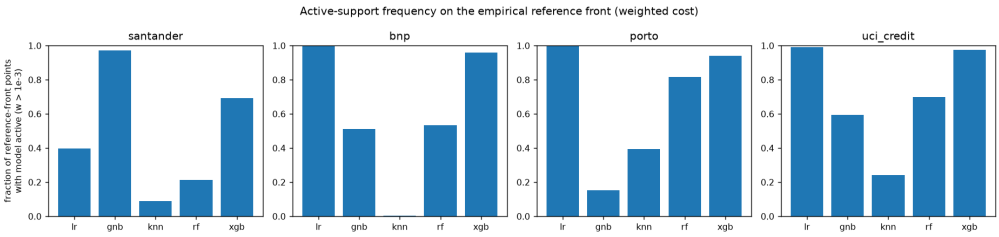


Figure S10: Frequency with which each support set appears on the empirical reference front.

Table S7: Complete paired primary effects: both cost definitions, all four comparisons, both endpoints, with means, medians, dispersion, both bootstrap intervals, win/tie/loss, Wilson interval, rank-biserial correlation and both tests. Showing 15 of 30 columns; the rest are in the source CSV.

cost	dataset	comparison	endpoint	n	mean	median	ci95 median lo	ci95 median hi	wins	ties	losses	rank biserial	p nb	p wilcoxon
weighted	santander	nbi_B vs nbi_A	igd_plus	30	0.120000	0.125100	0.086300	0.144900	30	0	0	1.000000	0.001500	0.000000
weighted	santander	nbi_B vs nbi_A	hv_ratio	30	0.179200	0.196000	0.142700	0.215900	30	0	0	1.000000	0.000600	0.000000
weighted	santander	nbi_C vs nbi_B	igd_plus	30	0.001600	0.001700	0.001100	0.002000	28	0	2	0.840900	0.049400	0.000000
weighted	santander	nbi_C vs nbi_B	hv_ratio	30	0.007700	0.008400	0.007200	0.009300	28	0	2	0.978500	0.000400	0.000000
weighted	santander	nbi_C vs ws_random_scalarization	igd_plus	30	-0.002900	-0.001800	-0.005400	0.000200	10	0	20	-0.471000	0.345200	0.023400
weighted	santander	nbi_C vs ws_random_scalarization	hv_ratio	30	0.010600	0.010400	0.008400	0.014100	25	0	5	0.931200	0.020500	0.000000
weighted	santander	nbi_C vs random_dirichlet_budget	igd_plus	30	0.819600	0.710200	0.355800	1.101600	30	0	0	1.000000	0.013000	0.000000
weighted	santander	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.709600	0.778500	0.510500	0.969900	30	0	0	1.000000	0.000100	0.000000
weighted	bnp	nbi_B vs nbi_A	igd_plus	30	0.238800	0.014600	0.013600	0.017200	30	0	0	1.000000	0.245900	0.000000
weighted	bnp	nbi_B vs nbi_A	hv_ratio	30	0.279200	0.058500	0.050600	0.070100	30	0	0	1.000000	0.165300	0.000000
weighted	bnp	nbi_C vs nbi_B	igd_plus	30	-0.002900	-0.002800	-0.004400	-0.001500	6	0	24	-0.741900	0.270700	0.000200
weighted	bnp	nbi_C vs nbi_B	hv_ratio	30	0.008600	0.010600	0.008700	0.012800	24	0	6	0.681700	0.211600	0.000700
weighted	bnp	nbi_C vs ws_random_scalarization	igd_plus	30	0.397500	0.292600	0.281300	0.314400	30	0	0	1.000000	0.035400	0.000000
weighted	bnp	nbi_C vs ws_random_scalarization	hv_ratio	30	0.538900	0.509900	0.494900	0.521700	30	0	0	1.000000	0.000000	0.000000
weighted	bnp	nbi_C vs random_dirichlet_budget	igd_plus	30	1.513500	1.512000	1.216100	1.751600	30	0	0	1.000000	0.000100	0.000000
weighted	bnp	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.942800	0.980500	0.976100	0.982900	30	0	0	1.000000	0.000000	0.000000
weighted	porto	nbi_B vs nbi_A	igd_plus	30	0.038000	0.029600	0.019300	0.073000	23	0	7	0.651600	0.238600	0.001200
weighted	porto	nbi_B vs nbi_A	hv_ratio	30	0.085300	0.075600	0.046200	0.157400	24	0	6	0.655900	0.246700	0.001100
weighted	porto	nbi_C vs nbi_B	igd_plus	30	0.042000	0.024400	0.016800	0.040800	28	0	2	0.896800	0.168700	0.000000
weighted	porto	nbi_C vs nbi_B	hv_ratio	30	0.108300	0.074000	0.039300	0.100400	28	0	2	0.909700	0.157200	0.000000
weighted	porto	nbi_C vs ws_random_scalarization	igd_plus	30	1.640600	0.090600	0.059300	0.103300	30	0	0	1.000000	0.711100	0.000000
weighted	porto	nbi_C vs ws_random_scalarization	hv_ratio	30	0.269800	0.217100	0.164500	0.248200	30	0	0	1.000000	0.058100	0.000000
weighted	porto	nbi_C vs random_dirichlet_budget	igd_plus	30	1.943100	1.789000	1.266000	2.276300	30	0	0	1.000000	0.003300	0.000000
weighted	porto	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.930000	0.980400	0.970900	0.984500	30	0	0	1.000000	0.000000	0.000000
weighted	uci_credit	nbi_B vs nbi_A	igd_plus	30	0.036200	0.049900	0.032700	0.060600	24	0	6	0.754800	0.113100	0.000100
weighted	uci_credit	nbi_B vs nbi_A	hv_ratio	30	0.059700	0.091600	0.038900	0.099800	24	1	5	0.746200	0.117000	0.000200
weighted	uci_credit	nbi_C vs nbi_B	igd_plus	30	0.158200	0.144600	0.126400	0.182200	29	0	1	0.995700	0.002000	0.000000
weighted	uci_credit	nbi_C vs nbi_B	hv_ratio	30	0.277400	0.275000	0.218700	0.329500	30	0	0	1.000000	0.001600	0.000000
weighted	uci_credit	nbi_C vs ws_random_scalarization	igd_plus	30	0.000800	0.000500	-0.001000	0.001100	18	0	12	0.122600	0.812800	0.569900
weighted	uci_credit	nbi_C vs ws_random_scalarization	hv_ratio	30	0.001500	0.000700	-0.003000	0.003100	16	0	14	0.058100	0.844100	0.792200
weighted	uci_credit	nbi_C vs random_dirichlet_budget	igd_plus	30	0.140300	0.140100	0.120900	0.151000	30	0	0	1.000000	0.000000	0.000000
weighted	uci_credit	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.251900	0.241000	0.224600	0.283600	30	0	0	1.000000	0.000000	0.000000
support	santander	nbi_B vs nbi_A	igd_plus	30	0.059900	0.058900	0.055300	0.065900	30	0	0	1.000000	0.000000	0.000000
support	santander	nbi_B vs nbi_A	hv_ratio	30	0.125500	0.130300	0.123600	0.141700	30	0	0	1.000000	0.000000	0.000000
support	santander	nbi_C vs nbi_B	igd_plus	30	0.006600	0.006400	0.006100	0.007200	30	0	0	1.000000	0.000000	0.000000
support	santander	nbi_C vs nbi_B	hv_ratio	30	0.005700	0.006500	0.002000	0.008600	28	0	2	0.909700	0.040500	0.000000
support	santander	nbi_C vs ws_random_scalarization	igd_plus	30	0.008700	0.008000	0.007300	0.008500	30	0	0	1.000000	0.000100	0.000000
support	santander	nbi_C vs ws_random_scalarization	hv_ratio	30	0.008600	0.008900	0.007300	0.010300	28	0	2	0.982800	0.004300	0.000000
support	santander	nbi_C vs random_dirichlet_budget	igd_plus	30	0.635700	0.661000	0.598200	0.730800	30	0	0	1.000000	0.000000	0.000000
support	santander	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.800900	0.915000	0.905700	0.919400	30	0	0	1.000000	0.000000	0.000000
support	bnp	nbi_B vs nbi_A	igd_plus	30	0.224700	0.018700	-0.003500	0.032600	19	0	11	0.402200	0.336400	0.054900
support	bnp	nbi_B vs nbi_A	hv_ratio	30	0.151000	0.074100	-0.016600	0.098200	20	0	10	0.268800	0.467400	0.205400
support	bnp	nbi_C vs nbi_B	igd_plus	30	-0.021000	-0.002500	-0.013600	-0.000100	9	1	20	-0.561300	0.410000	0.006200
support	bnp	nbi_C vs nbi_B	hv_ratio	30	-0.054700	-0.010900	-0.031000	-0.001600	5	1	24	-0.724700	0.288600	0.000300
support	bnp	nbi_C vs ws_random_scalarization	igd_plus	30	0.360500	0.266900	0.206000	0.340500	28	0	2	0.909700	0.113100	0.000000
support	bnp	nbi_C vs ws_random_scalarization	hv_ratio	30	0.320200	0.441800	0.125700	0.463600	23	0	7	0.840900	0.043900	0.000000
support	bnp	nbi_C vs random_dirichlet_budget	igd_plus	30	4.700800	0.960000	0.943700	9.017800	30	0	0	1.000000	0.087600	0.000000
support	bnp	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.735900	0.727600	0.574500	0.890500	30	0	0	1.000000	0.000000	0.000000
support	porto	nbi_B vs nbi_A	igd_plus	30	0.080200	0.063300	0.052800	0.098900	29	0	1	0.995700	0.017300	0.000000
support	porto	nbi_B vs nbi_A	hv_ratio	30	0.004000	0.005100	-0.046700	0.030100	19	0	11	0.040900	0.971600	0.855300
support	porto	nbi_C vs nbi_B	igd_plus	30	0.012300	0.006100	0.003600	0.011400	26	0	4	0.763400	0.396700	0.000100

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cost	dataset	comparison	endpoint	n	mean	median	ci95 median lo	ci95 median hi	wins	ties	losses	rank biserial	p nb	p wilcoxon
support	porto	nbi_C vs nbi_B	hv_ratio	30	0.098400	0.066200	0.025300	0.104600	23	0	7	0.720400	0.291700	0.000300
support	porto	nbi_C vs ws_random_scalarization	igd_plus	30	1.568000	0.085300	0.066500	0.121700	30	0	0	1.000000	0.705100	0.000000
support	porto	nbi_C vs ws_random_scalarization	hv_ratio	30	0.165600	0.092800	0.052400	0.137200	24	0	6	0.789200	0.290800	0.000000
support	porto	nbi_C vs random_dirichlet_budget	igd_plus	30	1.359000	1.222800	0.961800	1.446300	30	0	0	1.000000	0.000200	0.000000
support	porto	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.905000	0.920600	0.890600	0.958000	30	0	0	1.000000	0.000000	0.000000
support	uci_credit	nbi_B vs nbi_A	igd_plus	30	-0.023700	-0.010300	-0.013500	-0.007300	3	0	27	-0.638700	0.646500	0.001600
support	uci_credit	nbi_B vs nbi_A	hv_ratio	30	-0.042600	-0.015500	-0.022900	-0.013400	4	0	26	-0.759100	0.584000	0.000100
support	uci_credit	nbi_C vs nbi_B	igd_plus	30	0.030500	0.009700	0.006700	0.019900	26	0	4	0.754800	0.543400	0.000100
support	uci_credit	nbi_C vs nbi_B	hv_ratio	30	0.071700	0.052200	0.038300	0.059100	27	0	3	0.965600	0.351100	0.000000
support	uci_credit	nbi_C vs ws_random_scalarization	igd_plus	30	-0.027800	-0.030400	-0.035500	-0.021700	0	0	30	-1.000000	0.001600	0.000000
support	uci_credit	nbi_C vs ws_random_scalarization	hv_ratio	30	-0.038100	-0.039100	-0.046000	-0.031000	0	0	30	-1.000000	0.000000	0.000000
support	uci_credit	nbi_C vs random_dirichlet_budget	igd_plus	30	0.472000	0.512800	0.485500	0.560600	30	0	0	1.000000	0.000000	0.000000
support	uci_credit	nbi_C vs random_dirichlet_budget	hv_ratio	30	0.744000	0.840900	0.802300	0.850200	30	0	0	1.000000	0.000000	0.000000

Table S8: Holm-corrected test families, both cost definitions. Showing 11 of 16 columns; the rest are in the source CSV.

cost	dataset	comparison	endpoint	mean delta	t nadeau bengio	p nadeau bengio	p nb holm family dataset	p wilcoxon holm	rank biserial	win frac
weighted	santander	nbi_B vs nbi_A	igd_plus	0.120000	3.512200	0.001500	0.007400	0.000000	1.000000	1.000000
weighted	santander	nbi_B vs nbi_A	hv_ratio	0.179200	3.842600	0.000600	0.003700	0.000000	1.000000	1.000000
weighted	santander	nbi_C vs nbi_B	igd_plus	0.001600	2.050700	0.049400	0.098900	0.000000	0.840900	0.933300
weighted	santander	nbi_C vs nbi_B	hv_ratio	0.007700	4.046500	0.000400	0.002500	0.000000	0.978500	0.933300
weighted	santander	nbi_C vs ws_random_scalarization	igd_plus	-0.002900	-0.959700	0.345200	0.345200	0.023400	-0.471000	0.333300
weighted	santander	nbi_C vs ws_random_scalarization	hv_ratio	0.010600	2.451200	0.020500	0.061500	0.000000	0.931200	0.833300
weighted	santander	nbi_C vs random_dirichlet_budget	igd_plus	0.819600	2.647100	0.013000	0.051900	0.000000	1.000000	1.000000
weighted	santander	nbi_C vs random_dirichlet_budget	hv_ratio	0.709600	4.731400	0.000100	0.000400	0.000000	1.000000	1.000000
weighted	bnp	nbi_B vs nbi_A	igd_plus	0.238800	1.184300	0.245900	0.661100	0.000000	1.000000	1.000000
weighted	bnp	nbi_B vs nbi_A	hv_ratio	0.279200	1.423400	0.165300	0.661100	0.000000	1.000000	1.000000
weighted	bnp	nbi_C vs nbi_B	igd_plus	-0.002900	-1.123000	0.270700	0.661100	0.000300	-0.741900	0.200000
weighted	bnp	nbi_C vs nbi_B	hv_ratio	0.008600	1.277400	0.211600	0.661100	0.000700	0.681700	0.800000
weighted	bnp	nbi_C vs ws_random_scalarization	igd_plus	0.397500	2.206800	0.035400	0.177000	0.000000	1.000000	1.000000
weighted	bnp	nbi_C vs ws_random_scalarization	hv_ratio	0.538900	5.266600	0.000000	0.000100	0.000000	1.000000	1.000000
weighted	bnp	nbi_C vs random_dirichlet_budget	igd_plus	1.513500	4.375700	0.000100	0.000900	0.000000	1.000000	1.000000
weighted	bnp	nbi_C vs random_dirichlet_budget	hv_ratio	0.942800	22.052800	0.000000	0.000000	0.000000	1.000000	1.000000
weighted	porto	nbi_B vs nbi_A	igd_plus	0.038000	1.203200	0.238600	0.785800	0.002300	0.651600	0.766700
weighted	porto	nbi_B vs nbi_A	hv_ratio	0.085300	1.182400	0.246700	0.785800	0.002300	0.655900	0.800000
weighted	porto	nbi_C vs nbi_B	igd_plus	0.042000	1.411600	0.168700	0.785800	0.000000	0.896800	0.933300
weighted	porto	nbi_C vs nbi_B	hv_ratio	0.108300	1.452200	0.157200	0.785800	0.000000	0.909700	0.933300
weighted	porto	nbi_C vs ws_random_scalarization	igd_plus	1.640600	0.374000	0.711100	0.785800	0.000000	1.000000	1.000000
weighted	porto	nbi_C vs ws_random_scalarization	hv_ratio	0.269800	1.972700	0.058100	0.348800	0.000000	1.000000	1.000000
weighted	porto	nbi_C vs random_dirichlet_budget	igd_plus	1.943100	3.198300	0.003300	0.023300	0.000000	1.000000	1.000000
weighted	porto	nbi_C vs random_dirichlet_budget	hv_ratio	0.930000	15.058800	0.000000	0.000000	0.000000	1.000000	1.000000
weighted	uci_credit	nbi_B vs nbi_A	igd_plus	0.036200	1.634000	0.113100	0.452300	0.000500	0.754800	0.800000
weighted	uci_credit	nbi_B vs nbi_A	hv_ratio	0.059700	1.615500	0.117000	0.452300	0.000500	0.746200	0.800000
weighted	uci_credit	nbi_C vs nbi_B	igd_plus	0.158200	3.404800	0.002000	0.009800	0.000000	0.995700	0.966700
weighted	uci_credit	nbi_C vs nbi_B	hv_ratio	0.277400	3.484100	0.001600	0.009500	0.000000	1.000000	1.000000
weighted	uci_credit	nbi_C vs ws_random_scalarization	igd_plus	0.000800	0.238900	0.812800	1.000000	1.000000	0.122600	0.600000
weighted	uci_credit	nbi_C vs ws_random_scalarization	hv_ratio	0.001500	0.198400	0.844100	1.000000	1.000000	0.058100	0.533300
weighted	uci_credit	nbi_C vs random_dirichlet_budget	igd_plus	0.140300	6.535400	0.000000	0.000000	0.000000	1.000000	1.000000
weighted	uci_credit	nbi_C vs random_dirichlet_budget	hv_ratio	0.251900	8.166300	0.000000	0.000000	0.000000	1.000000	1.000000
support	santander	nbi_B vs nbi_A	igd_plus	0.059900	5.725700	0.000000	0.000000	0.000000	1.000000	1.000000
support	santander	nbi_B vs nbi_A	hv_ratio	0.125500	7.298600	0.000000	0.000000	0.000000	1.000000	1.000000
support	santander	nbi_C vs nbi_B	igd_plus	0.006600	7.805900	0.000000	0.000000	0.000000	1.000000	1.000000
support	santander	nbi_C vs nbi_B	hv_ratio	0.005700	2.144300	0.040500	0.040500	0.000000	0.909700	0.933300
support	santander	nbi_C vs ws_random_scalarization	igd_plus	0.008700	4.375500	0.000100	0.000400	0.000000	1.000000	1.000000
support	santander	nbi_C vs ws_random_scalarization	hv_ratio	0.008600	3.102300	0.004300	0.008500	0.000000	0.982800	0.933300
support	santander	nbi_C vs random_dirichlet_budget	igd_plus	0.635700	5.197000	0.000000	0.000100	0.000000	1.000000	1.000000
support	santander	nbi_C vs random_dirichlet_budget	hv_ratio	0.800900	6.340200	0.000000	0.000000	0.000000	1.000000	1.000000
support	bnp	nbi_B vs nbi_A	igd_plus	0.224700	0.977500	0.336400	1.000000	0.109800	0.402200	0.633300
support	bnp	nbi_B vs nbi_A	hv_ratio	0.151000	0.736400	0.467400	1.000000	0.205400	0.268800	0.666700
support	bnp	nbi_C vs nbi_B	igd_plus	-0.021000	-0.836100	0.410000	1.000000	0.018600	-0.561300	0.300000
support	bnp	nbi_C vs nbi_B	hv_ratio	-0.054700	-1.081000	0.288600	1.000000	0.001000	-0.724700	0.166700
support	bnp	nbi_C vs ws_random_scalarization	igd_plus	0.360500	1.633700	0.113100	0.565700	0.000000	0.909700	0.933300
support	bnp	nbi_C vs ws_random_scalarization	hv_ratio	0.320200	2.106500	0.043900	0.307500	0.000100	0.840900	0.766700
support	bnp	nbi_C vs random_dirichlet_budget	igd_plus	4.700800	1.767800	0.087600	0.525700	0.000000	1.000000	1.000000
support	bnp	nbi_C vs random_dirichlet_budget	hv_ratio	0.735900	8.561800	0.000000	0.000000	0.000000	1.000000	1.000000
support	porto	nbi_B vs nbi_A	igd_plus	0.080200	2.525000	0.017300	0.103800	0.000000	0.995700	0.966700
support	porto	nbi_B vs nbi_A	hv_ratio	0.004000	0.035900	0.971600	1.000000	0.855300	0.040900	0.633300
support	porto	nbi_C vs nbi_B	igd_plus	0.012300	0.860300	0.396700	1.000000	0.000300	0.763400	0.866700
support	porto	nbi_C vs nbi_B	hv_ratio	0.098400	1.073900	0.291700	1.000000	0.000600	0.720400	0.766700
support	porto	nbi_C vs ws_random_scalarization	igd_plus	1.568000	0.382200	0.705100	1.000000	0.000000	1.000000	1.000000

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cost	dataset	comparison	endpoint	mean delta	t nadeau bengio	p nadeau bengio	p nb holm family dataset	p wilcoxon holm	rank biserial	win frac
support	porto	nbi_C vs ws_random_scalarization	hv_ratio	0.165600	1.075900	0.290800	1.000000	0.000200	0.789200	0.800000
support	porto	nbi_C vs random_dirichlet_budget	igd_plus	1.359000	4.269700	0.000200	0.001300	0.000000	1.000000	1.000000
support	porto	nbi_C vs random_dirichlet_budget	hv_ratio	0.905000	21.029700	0.000000	0.000000	0.000000	1.000000	1.000000
support	uci_credit	nbi_B vs nbi_A	igd_plus	-0.023700	-0.463500	0.646500	1.000000	0.001600	-0.638700	0.100000
support	uci_credit	nbi_B vs nbi_A	hv_ratio	-0.042600	-0.553800	0.584000	1.000000	0.000300	-0.759100	0.133300
support	uci_credit	nbi_C vs nbi_B	igd_plus	0.030500	0.614900	0.543400	1.000000	0.000300	0.754800	0.866700
support	uci_credit	nbi_C vs nbi_B	hv_ratio	0.071700	0.947700	0.351100	1.000000	0.000000	0.965600	0.900000
support	uci_credit	nbi_C vs ws_random_scalarization	igd_plus	-0.027800	-3.470200	0.001600	0.008200	0.000000	-1.000000	0.000000
support	uci_credit	nbi_C vs ws_random_scalarization	hv_ratio	-0.038100	-5.189200	0.000000	0.000100	0.000000	-1.000000	0.000000
support	uci_credit	nbi_C vs random_dirichlet_budget	igd_plus	0.472000	5.498800	0.000000	0.000000	0.000000	1.000000	1.000000
support	uci_credit	nbi_C vs random_dirichlet_budget	hv_ratio	0.744000	6.991600	0.000000	0.000000	0.000000	1.000000	1.000000

Table S9: Wilson and Jeffreys intervals for every reported frequency. First 70 of 140 rows; the complete table is the source CSV.

quantity	dataset	level	k	n	p	wilson lo	wilson hi	jeffreys lo	jeffreys hi
reliability_gate_pass	santander	roc_auc	0	30	0.000000	0.000000	0.113500	0.000000	0.079700
reliability_gate_pass	santander	log_loss	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
reliability_gate_pass	santander	brier	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
reliability_gate_pass	santander	pr_auc	0	30	0.000000	0.000000	0.113500	0.000000	0.079700
reliability_gate_pass	bnp	roc_auc	5	30	0.166700	0.073400	0.335600	0.066600	0.327400
reliability_gate_pass	bnp	log_loss	3	30	0.100000	0.034600	0.256200	0.029000	0.243400
reliability_gate_pass	bnp	brier	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
reliability_gate_pass	bnp	pr_auc	13	30	0.433300	0.273800	0.608000	0.268900	0.609900
reliability_gate_pass	porto	roc_auc	16	30	0.533300	0.361400	0.697700	0.358700	0.701900
reliability_gate_pass	porto	log_loss	12	30	0.400000	0.245900	0.576800	0.240500	0.577800
reliability_gate_pass	porto	brier	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
reliability_gate_pass	porto	pr_auc	27	30	0.900000	0.743800	0.965400	0.756600	0.971000
reliability_gate_pass	uci_credit	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
reliability_gate_pass	uci_credit	log_loss	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
reliability_gate_pass	uci_credit	brier	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
reliability_gate_pass	uci_credit	pr_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_B vs nbi_A, igd_plus]	santander	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_B vs nbi_A, hv_ratio]	santander	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs nbi_B, igd_plus]	santander	weighted	28	30	0.933300	0.786800	0.981500	0.802900	0.985900
win_frac[nbi_C vs nbi_B, hv_ratio]	santander	weighted	28	30	0.933300	0.786800	0.981500	0.802900	0.985900
win_frac[nbi_C vs ws_random_scalarization, igd_plus]	santander	weighted	10	30	0.333300	0.192300	0.512200	0.186000	0.511100
win_frac[nbi_C vs ws_random_scalarization, hv_ratio]	santander	weighted	25	30	0.833300	0.664400	0.926600	0.672600	0.933400
win_frac[nbi_C vs random_dirichlet_budget, igd_plus]	santander	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs random_dirichlet_budget, hv_ratio]	santander	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_B vs nbi_A, igd_plus]	bnp	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_B vs nbi_A, hv_ratio]	bnp	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs nbi_B, igd_plus]	bnp	weighted	6	30	0.200000	0.095100	0.373100	0.088100	0.366600
win_frac[nbi_C vs nbi_B, hv_ratio]	bnp	weighted	24	30	0.800000	0.626900	0.904900	0.633400	0.911900
win_frac[nbi_C vs ws_random_scalarization, igd_plus]	bnp	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs ws_random_scalarization, hv_ratio]	bnp	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs random_dirichlet_budget, igd_plus]	bnp	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs random_dirichlet_budget, hv_ratio]	bnp	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_B vs nbi_A, igd_plus]	porto	weighted	23	30	0.766700	0.590700	0.882100	0.595600	0.889100
win_frac[nbi_B vs nbi_A, hv_ratio]	porto	weighted	24	30	0.800000	0.626900	0.904900	0.633400	0.911900
win_frac[nbi_C vs nbi_B, igd_plus]	porto	weighted	28	30	0.933300	0.786800	0.981500	0.802900	0.985900
win_frac[nbi_C vs nbi_B, hv_ratio]	porto	weighted	28	30	0.933300	0.786800	0.981500	0.802900	0.985900
win_frac[nbi_C vs ws_random_scalarization, igd_plus]	porto	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs ws_random_scalarization, hv_ratio]	porto	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs random_dirichlet_budget, igd_plus]	porto	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs random_dirichlet_budget, hv_ratio]	porto	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_B vs nbi_A, igd_plus]	uci_credit	weighted	24	30	0.800000	0.626900	0.904900	0.633400	0.911900
win_frac[nbi_B vs nbi_A, hv_ratio]	uci_credit	weighted	24	30	0.800000	0.626900	0.904900	0.633400	0.911900
win_frac[nbi_C vs nbi_B, igd_plus]	uci_credit	weighted	29	30	0.966700	0.833300	0.994100	0.854600	0.996400
win_frac[nbi_C vs nbi_B, hv_ratio]	uci_credit	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs ws_random_scalarization, igd_plus]	uci_credit	weighted	18	30	0.600000	0.423200	0.754100	0.422200	0.759500
win_frac[nbi_C vs ws_random_scalarization, hv_ratio]	uci_credit	weighted	16	30	0.533300	0.361400	0.697700	0.358700	0.701900
win_frac[nbi_C vs random_dirichlet_budget, igd_plus]	uci_credit	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
win_frac[nbi_C vs random_dirichlet_budget, hv_ratio]	uci_credit	weighted	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
nbi_A_feasibility_rate>=0.9	santander	A	26	30	0.866700	0.703200	0.946900	0.713500	0.953300
nbi_B_feasibility_rate>=0.9	santander	B	23	30	0.766700	0.590700	0.882100	0.595600	0.889100
nbi_C_feasibility_rate>=0.9	santander	C	0	30	0.000000	0.000000	0.113500	0.000000	0.079700
nbi_A_feasibility_rate>=0.9	bnp	A	27	30	0.900000	0.743800	0.965400	0.756600	0.971000
nbi_B_feasibility_rate>=0.9	bnp	B	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
nbi_C_feasibility_rate>=0.9	bnp	C	2	30	0.066700	0.018500	0.213200	0.014100	0.197100
nbi_A_feasibility_rate>=0.9	porto	A	10	30	0.333300	0.192300	0.512200	0.186000	0.511100
nbi_B_feasibility_rate>=0.9	porto	B	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
nbi_C_feasibility_rate>=0.9	porto	C	29	30	0.966700	0.833300	0.994100	0.854600	0.996400
nbi_A_feasibility_rate>=0.9	uci_credit	A	3	30	0.100000	0.034600	0.256200	0.029000	0.243400
nbi_B_feasibility_rate>=0.9	uci_credit	B	3	30	0.100000	0.034600	0.256200	0.029000	0.243400
nbi_C_feasibility_rate>=0.9	uci_credit	C	29	30	0.966700	0.833300	0.994100	0.854600	0.996400
beta_sign_positive[gnb*knn, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[gnb*rf, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[gnb*xgb, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[knn*rf, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[knn*xgb, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[lr*gnb, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[lr*knn, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[lr*rf, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[lr*xgb, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000
beta_sign_positive[rf*xgb, roc_auc]	santander	roc_auc	30	30	1.000000	0.886500	1.000000	0.920300	1.000000

Sample-core repetition and the descriptive effect sizes. Scored against the sampled core, which contains no output of the five compared methods, rather than the augmented reference, the NBI-B versus NBI-A hypervolume median falls from +0.196 to +0.145 on Santander and rises from +0.058 to +0.070 on BNP Paribas, and the ordering of the four datasets by median gain stays inverted with respect to surrogate reliability (+0.145, +0.089, +0.078 and +0.070). For NBI-C versus NBI-B on BNP Paribas the medians are +0.0106 hypervolume and −0.0028 IGD^+ against the augmented reference, and +0.0077 and +0.0025 against the sample core, which is why the main text records that split as reference-dependent. The same holds for NBI-C versus random

scalarization on Santander: NBI-C wins 25/30 partitions on hypervolume and 10/30 on IGD^+ against the augmented reference, and its IGD^+ deficit does not persist against the sample core or against the augmented union with NSGA-II. The NBI-C versus NBI-B means are +0.108 on Porto Seguro and +0.277 on UCI credit, against anchor-effect means of +0.085 and +0.060 on the same datasets. Where the corrected test is uninformative the three descriptive quantities reported in its place are the median with its bootstrap interval, the win fraction with its Wilson interval (for example $[0.89, 1.00]$ for a 30/30 fraction) and the rank-biserial correlation (1.00 on the two unanimous datasets, 0.65–0.75 on the other two). None of the three corrects for partition overlap, so none establishes significance.

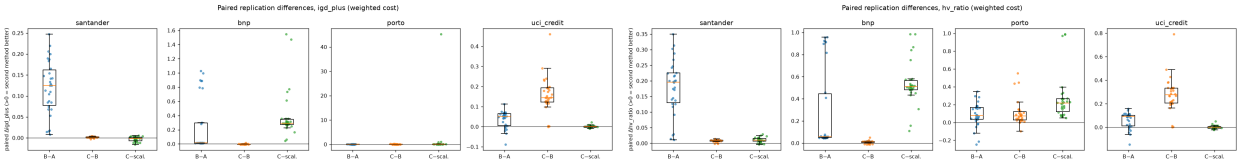


Figure S11: Paired differences on both primary endpoints, all comparisons.

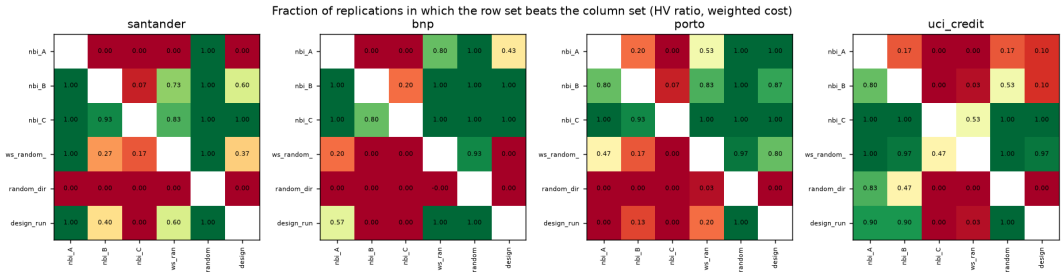


Figure S12: Win/tie/loss heat map over all pairwise set comparisons.

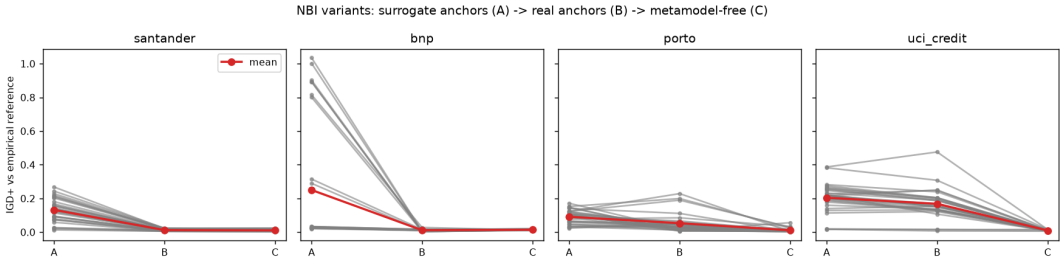


Figure S13: Paired comparison of the three NBI variants.

S6 Reliability gate and regime analysis

Table S10: Reliability gate for all four responses, including the Brier and PR-AUC responses omitted from the main text. The interval columns are the percentile-bootstrap 95% interval of the mean external R^2, printed beside its median. Showing 11 of 15 columns; the rest are in the source CSV.

dataset	response	n	pass	p pass	wilson lo	wilson hi	r2 external median	r2 external ci95 lo	r2 external ci95 hi	spearman median
santander	roc_auc	30	0	0.000000	0.000000	0.113500	-0.333200	-0.558500	-0.199000	0.830200
santander	log_loss	30	30	1.000000	0.886500	1.000000	0.970900	0.967300	0.972500	0.993900
santander	brier	30	30	1.000000	0.886500	1.000000	1.000000	1.000000	1.000000	1.000000
santander	pr_auc	30	0	0.000000	0.000000	0.113500	-0.221600	-0.578400	-0.166800	0.760100

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dataset	response	n	pass	p pass	wilson lo	wilson hi	r2 external median	r2 external ci95 lo	r2 external ci95 hi	spearman median
bnp	roc_auc	30	5	0.166700	0.073400	0.335600	0.117700	-0.034500	0.244100	0.908300
bnp	log_loss	30	3	0.100000	0.034600	0.256200	-0.180200	-0.245900	0.041800	0.955500
bnp	brier	30	30	1.000000	0.886500	1.000000	1.000000	1.000000	1.000000	1.000000
bnp	pr_auc	30	13	0.433300	0.273800	0.608000	0.460800	0.432300	0.551200	0.929500
porto	roc_auc	30	16	0.533300	0.361400	0.697700	0.519000	0.482200	0.580500	0.924000
porto	log_loss	30	12	0.400000	0.245900	0.576800	0.505600	0.492200	0.579800	0.994700
porto	brier	30	30	1.000000	0.886500	1.000000	1.000000	1.000000	1.000000	1.000000
porto	pr_auc	30	27	0.900000	0.743800	0.965400	0.779300	0.731800	0.801600	0.938200
uci_credit	roc_auc	30	30	1.000000	0.886500	1.000000	0.991500	0.989200	0.992100	0.994800
uci_credit	log_loss	30	30	1.000000	0.886500	1.000000	0.992600	0.990500	0.993200	0.992700
uci_credit	brier	30	30	1.000000	0.886500	1.000000	1.000000	1.000000	1.000000	1.000000
uci_credit	pr_auc	30	30	1.000000	0.886500	1.000000	0.929900	0.901100	0.931600	0.984000

Table S11: Anchor gain split by gate outcome, separately for each gate response. On BNP Paribas the ROC-AUC split restates NBI-A's collapse and must not be read as evidence that real anchors add more when the ROC-AUC surface is reliable. First 40 of 64 rows; the complete table is the source CSV. Showing 12 of 16 columns; the rest are in the source CSV.

dataset	gate response	gate passed	cost	endpoint	comparison	n	mean	median	wins	ties	losses
santander	roc_auc	True	weighted	igd_plus	nbi_B vs nbi_A	0	–	–	–	–	–
santander	roc_auc	False	weighted	igd_plus	nbi_B vs nbi_A	30	0.120000	0.125100	30.000000	0.000000	0.000000
santander	roc_auc	True	weighted	hv_ratio	nbi_B vs nbi_A	0	–	–	–	–	–
santander	roc_auc	False	weighted	hv_ratio	nbi_B vs nbi_A	30	0.179200	0.196000	30.000000	0.000000	0.000000
santander	roc_auc	True	support	igd_plus	nbi_B vs nbi_A	0	–	–	–	–	–
santander	roc_auc	False	support	igd_plus	nbi_B vs nbi_A	30	0.059900	0.058900	30.000000	0.000000	0.000000
santander	roc_auc	True	support	hv_ratio	nbi_B vs nbi_A	0	–	–	–	–	–
santander	roc_auc	False	support	hv_ratio	nbi_B vs nbi_A	30	0.125500	0.130300	30.000000	0.000000	0.000000
santander	log_loss	True	weighted	igd_plus	nbi_B vs nbi_A	30	0.120000	0.125100	30.000000	0.000000	0.000000
santander	log_loss	False	weighted	igd_plus	nbi_B vs nbi_A	0	–	–	–	–	–
santander	log_loss	True	weighted	hv_ratio	nbi_B vs nbi_A	30	0.179200	0.196000	30.000000	0.000000	0.000000
santander	log_loss	False	weighted	hv_ratio	nbi_B vs nbi_A	0	–	–	–	–	–
santander	log_loss	True	support	igd_plus	nbi_B vs nbi_A	30	0.059900	0.058900	30.000000	0.000000	0.000000
santander	log_loss	False	support	igd_plus	nbi_B vs nbi_A	0	–	–	–	–	–
santander	log_loss	True	support	hv_ratio	nbi_B vs nbi_A	30	0.125500	0.130300	30.000000	0.000000	0.000000
santander	log_loss	False	support	hv_ratio	nbi_B vs nbi_A	0	–	–	–	–	–
bnp	roc_auc	True	weighted	igd_plus	nbi_B vs nbi_A	5	0.579200	0.796400	5.000000	0.000000	0.000000
bnp	roc_auc	False	weighted	igd_plus	nbi_B vs nbi_A	25	0.170700	0.014300	25.000000	0.000000	0.000000
bnp	roc_auc	True	weighted	hv_ratio	nbi_B vs nbi_A	5	0.638300	0.899000	5.000000	0.000000	0.000000
bnp	roc_auc	False	weighted	hv_ratio	nbi_B vs nbi_A	25	0.207400	0.056100	25.000000	0.000000	0.000000
bnp	roc_auc	True	support	igd_plus	nbi_B vs nbi_A	5	0.559500	0.866000	4.000000	0.000000	1.000000
bnp	roc_auc	False	support	igd_plus	nbi_B vs nbi_A	25	0.157800	0.016600	15.000000	0.000000	10.000000
bnp	roc_auc	True	support	hv_ratio	nbi_B vs nbi_A	5	0.470900	0.544800	5.000000	0.000000	0.000000
bnp	roc_auc	False	support	hv_ratio	nbi_B vs nbi_A	25	0.087000	0.072800	15.000000	0.000000	10.000000
bnp	log_loss	True	weighted	igd_plus	nbi_B vs nbi_A	3	0.013900	0.014300	3.000000	0.000000	0.000000
bnp	log_loss	False	weighted	igd_plus	nbi_B vs nbi_A	27	0.263800	0.015100	27.000000	0.000000	0.000000
bnp	log_loss	True	weighted	hv_ratio	nbi_B vs nbi_A	3	0.054200	0.052300	3.000000	0.000000	0.000000
bnp	log_loss	False	weighted	hv_ratio	nbi_B vs nbi_A	27	0.304200	0.058800	27.000000	0.000000	0.000000
bnp	log_loss	True	support	igd_plus	nbi_B vs nbi_A	3	0.025500	0.023200	3.000000	0.000000	0.000000
bnp	log_loss	False	support	igd_plus	nbi_B vs nbi_A	27	0.246900	0.016600	16.000000	0.000000	11.000000
bnp	log_loss	True	support	hv_ratio	nbi_B vs nbi_A	3	0.076300	0.075400	3.000000	0.000000	0.000000
bnp	log_loss	False	support	hv_ratio	nbi_B vs nbi_A	27	0.159300	0.071000	17.000000	0.000000	10.000000
porto	roc_auc	True	weighted	igd_plus	nbi_B vs nbi_A	16	0.032900	0.029600	11.000000	0.000000	5.000000
porto	roc_auc	False	weighted	igd_plus	nbi_B vs nbi_A	14	0.043800	0.030500	12.000000	0.000000	2.000000
porto	roc_auc	True	weighted	hv_ratio	nbi_B vs nbi_A	16	0.073400	0.068000	12.000000	0.000000	4.000000
porto	roc_auc	False	weighted	hv_ratio	nbi_B vs nbi_A	14	0.098800	0.075600	12.000000	0.000000	2.000000
porto	roc_auc	True	support	igd_plus	nbi_B vs nbi_A	16	0.085300	0.069500	16.000000	0.000000	0.000000
porto	roc_auc	False	support	igd_plus	nbi_B vs nbi_A	14	0.074300	0.058300	13.000000	0.000000	1.000000
porto	roc_auc	True	support	hv_ratio	nbi_B vs nbi_A	16	-0.037700	0.003300	9.000000	0.000000	7.000000
porto	roc_auc	False	support	hv_ratio	nbi_B vs nbi_A	14	0.051700	0.009400	10.000000	0.000000	4.000000

Table S12: Post hoc split-sample reading of the gate (added in revision). Recorded passes use the same 100 compositions to select the order and to judge it; the expected passes are sums over 30 partitions of the pass fraction over 400 half-sample evaluations, with the order selected and judged on the same half (same half) or selected on one half and judged on the other (cross half). Order agreement: share of half-sample selections equal to the recorded order, median over partitions. Difference: same half minus cross half. Verdicts differing: partitions whose cross-half majority verdict differs from the recorded one. Source: `scripts/pco213_gate_split_sample.py`.

dataset	response	recorded	same half	cross half	difference	verdicts differing	order agreement
santander	roc_auc	0	0.120000	0.010000	0.110000	0	0.733000
santander	log_loss	30	30.000000	30.000000	0.000000	0	0.965000
santander	brier	30	30.000000	30.000000	0.000000	0	1.000000

Continued on next page

dataset	response	recorded	same half	cross half	difference	verdicts differing	order agreement
santander	pr_auc	0	0.010000	0.000000	0.010000	0	0.802000
bnp	roc_auc	5	5.780000	4.230000	1.560000	1	0.751000
bnp	log_loss	3	3.550000	2.100000	1.450000	3	0.759000
bnp	brier	30	30.000000	30.000000	0.000000	0	1.000000
bnp	pr_auc	13	15.140000	14.320000	0.820000	1	0.835000
porto	roc_auc	16	14.970000	11.700000	3.270000	6	0.688000
porto	log_loss	12	11.120000	7.200000	3.920000	3	0.711000
porto	brier	30	30.000000	30.000000	0.000000	0	1.000000
porto	pr_auc	27	24.830000	23.980000	0.850000	3	0.934000
uci_credit	roc_auc	30	30.000000	30.000000	0.000000	0	0.846000
uci_credit	log_loss	30	30.000000	30.000000	0.000000	0	0.877000
uci_credit	brier	30	30.000000	30.000000	0.000000	0	1.000000
uci_credit	pr_auc	30	29.940000	29.740000	0.200000	0	0.968000

Table S13: Exploratory regime diagnostics: Sarle's bimodality coefficient and a one- versus two-component Gaussian-mixture BIC difference. No formal multimodality test is claimed. First 50 of 108 rows; the complete table is the source CSV. Showing 10 of 14 columns; the rest are in the source CSV.

dataset	set	cost	metric	n	mean	median	sd	bimodality coefficient	gmm bic delta
santander	nbi_A	weighted	hv_ratio	30	0.800400	0.788700	0.092200	0.342700	-9.597200
santander	nbi_A	weighted	igd_plus	30	0.132100	0.136200	0.069500	0.385900	-9.161700
santander	nbi_A	support	hv_ratio	30	0.862400	0.853100	0.029800	0.520500	2.073900
santander	nbi_A	support	igd_plus	30	0.068300	0.067200	0.018500	0.278000	-9.079600
santander	nbi_A	-	success_rate	30	0.975300	1.000000	0.064200	0.914800	225.906700
santander	nbi_A	-	anchor_auc_cost	30	45.730400	53.680300	18.463900	0.882900	101.826100
santander	nbi_A	-	anchor_ll_cost	30	26.583000	27.473900	3.167100	0.882400	63.674100
santander	nbi_B	weighted	hv_ratio	30	0.979600	0.980700	0.006600	0.489200	-0.676400
santander	nbi_B	weighted	igd_plus	30	0.012100	0.010500	0.005800	0.547200	1.070900
santander	nbi_B	support	hv_ratio	30	0.987800	0.986200	0.005600	0.476800	15.233400
santander	nbi_B	support	igd_plus	30	0.008400	0.008200	0.001700	0.238200	-10.118200
santander	nbi_B	-	success_rate	30	0.949000	0.984800	0.059700	0.571600	96.077600
santander	nbi_B	-	anchor_auc_cost	30	3.737100	2.371800	3.615300	0.709700	21.014200
santander	nbi_B	-	anchor_ll_cost	30	0.876000	0.875600	0.009600	0.392800	-5.980000
santander	nbi_C	weighted	hv_ratio	30	0.987300	0.989000	0.007300	0.578300	3.562000
santander	nbi_C	weighted	igd_plus	30	0.010500	0.009500	0.006000	0.524900	-0.306700
santander	nbi_C	support	hv_ratio	30	0.993500	0.995000	0.004600	0.693000	11.827600
santander	nbi_C	support	igd_plus	30	0.001800	0.001500	0.001000	0.610100	-10.151100
santander	nbi_C	-	success_rate	30	0.828300	0.833300	0.017500	0.481000	12.074200
santander	nbi_C	-	anchor_auc_cost	30	3.737100	2.371800	3.615300	0.709700	21.014200
santander	nbi_C	-	anchor_ll_cost	30	0.876000	0.875600	0.009600	0.392800	-5.980000
santander	nbi_A	weighted	spearman(hv_ratio, r2ext_auc)	30	-0.244000	–	–	–	–
santander	nbi_A	weighted	spearman(hv_ratio, r2ext_ll)	30	-0.049200	–	–	–	–
santander	nbi_A	weighted	spearman(hv_ratio, spearman_auc)	30	-0.395300	–	–	–	–
santander	nbi_A	weighted	spearman(hv_ratio, anchor_auc_cost_A)	30	-0.094500	–	–	–	–
santander	nbi_A	weighted	spearman(hv_ratio, anchor_ll_cost_A)	30	0.405100	–	–	–	–
santander	nbi_A	weighted	spearman(hv_ratio, split_prevalence)	30	–	–	–	–	–
bnp	nbi_A	weighted	hv_ratio	30	0.694500	0.913000	0.361900	0.840800	48.138100
bnp	nbi_A	weighted	igd_plus	30	0.250200	0.027100	0.376300	0.840600	50.243500
bnp	nbi_A	support	hv_ratio	30	0.714200	0.904300	0.359500	0.854200	53.356800
bnp	nbi_A	support	igd_plus	30	0.266000	0.040500	0.421600	0.846100	75.468400
bnp	nbi_A	-	success_rate	30	0.946000	1.000000	0.156800	0.839000	45.520000
bnp	nbi_A	-	anchor_auc_cost	30	7.914400	3.479400	9.303400	0.856600	130.868300
bnp	nbi_A	-	anchor_ll_cost	30	3.466900	3.473900	0.054700	0.385200	-7.950800
bnp	nbi_B	weighted	hv_ratio	30	0.973800	0.971000	0.013100	0.222900	6.452100
bnp	nbi_B	weighted	igd_plus	30	0.011400	0.012000	0.005100	0.225300	-3.101300
bnp	nbi_B	support	hv_ratio	30	0.865200	0.970600	0.149400	0.713400	62.403000
bnp	nbi_B	support	igd_plus	30	0.041300	0.018600	0.046500	0.709100	13.960700
bnp	nbi_B	-	success_rate	30	0.995500	1.000000	0.016000	0.862700	143.388800
bnp	nbi_B	-	anchor_auc_cost	30	4.373900	4.381200	0.070300	0.303100	-2.638400
bnp	nbi_B	-	anchor_ll_cost	30	3.835300	3.826800	0.061000	0.308400	-10.092100
bnp	nbi_C	weighted	hv_ratio	30	0.982400	0.982900	0.003300	0.327700	-9.365600
bnp	nbi_C	weighted	igd_plus	30	0.014200	0.013600	0.002600	0.461700	-7.783700
bnp	nbi_C	support	hv_ratio	30	0.810500	0.968600	0.195500	0.794200	87.136400
bnp	nbi_C	support	igd_plus	30	0.062200	0.029800	0.089000	0.781600	34.150700
bnp	nbi_C	-	success_rate	30	0.833800	0.833300	0.038300	0.319500	-5.490200
bnp	nbi_C	-	anchor_auc_cost	30	4.373900	4.381200	0.070300	0.303100	-2.638400
bnp	nbi_C	-	anchor_ll_cost	30	3.835300	3.826800	0.061000	0.308400	-10.092100
bnp	nbi_A	weighted	spearman(hv_ratio, r2ext_auc)	30	-0.311200	–	–	–	–
bnp	nbi_A	weighted	spearman(hv_ratio, r2ext_ll)	30	0.295200	–	–	–	–

Split-sample reading of the gate (post hoc). Order selection and the gate use the same 100 compositions, so a recorded pass is not an independent verdict on the selected order. Table S12 re-reads the gate from the stored records alone: the Scheffé surfaces are fitted on the design points only, so their predictions at the 100 compositions are fixed, and only the compositions used to select and to judge are varied. Each of 200 stratified halves (30 of the 60 Dirichlet($\mathbf{1}$) and 20 of the 40 Dirichlet($0.5 \cdot \mathbf{1}$) compositions, seed 20261013) selects the order with the frozen parsimony rule and judges it with the frozen gate on the other half, in both directions; selecting and judging on the same half is the control at equal sample size. The extreme verdicts do not move: UCI credit keeps a pass verdict in every partition on every response, Santander's ROC-AUC and PR-AUC surfaces almost never, and the Brier surfaces always. The intermediate cases are where reuse matters: from same half to cross half, the expected passes fall from 14.97 to 11.70 of 30 for Porto Seguro's ROC-AUC surface, from 11.12 to 7.20 for its log-loss surface, and from 5.78 to 4.23 and from 3.55 to 2.10 for BNP Paribas's ROC-AUC and log-loss surfaces; against the recorded counts, which also use all 100 compositions, the cross-half reading is lower in these cases and slightly higher for BNP Paribas's PR-AUC surface (14.32 against 13). The recorded gate counts, the selected orders and every NBI-A result are unchanged: NBI-A ran on the order selected with all 100 compositions, and its result under another order would need a new NBI run, which is not made.

The BNP Paribas collapse, in full. NBI-A's weighted hypervolume ratio splits into 21 partitions at 0.891–0.928, two intermediate, and seven collapsing to 0.036–0.120: partitions 7, 9, 18, 20, 23, 26 and 27. The seven coincide exactly with the partitions in which the parsimony rule selects the quadratic ROC-AUC order (7 of 7, against 0 of 22 linear-selected partitions). Their ROC-AUC external R^2 is *higher* (median 0.456 against 0.075) and their ROC-AUC surface passes the gate more often (3 of 7 against 2 of 23), which is the basis for the main text's statement that the ROC-AUC gate does not single out this failure. NBI-B reaches 0.935–0.996 in all thirty partitions. In six of the seven collapses the surrogate anchor carries k-nearest neighbours and costs 16–31 ms per 1,000 rows, giving Spearman -0.69 between anchor cost and hypervolume; partition 26 collapses with a cheap anchor, which is why anchor cost is described as a marker rather than the mechanism.

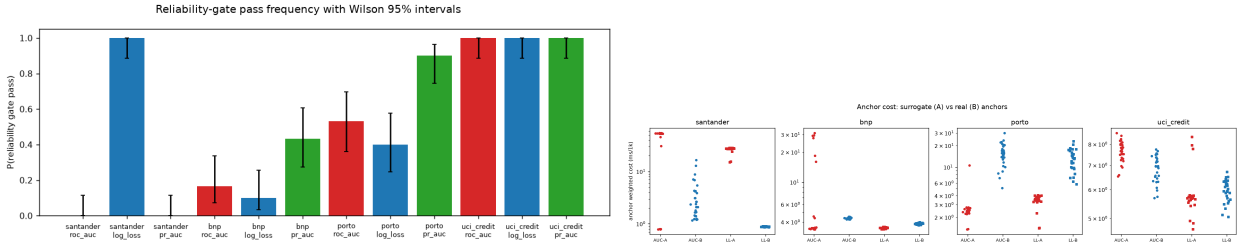


Figure S14: Gate pass frequencies and the distribution of anchor costs.

S7 NBI solver diagnostics

Table S14: NBI solver diagnostics per dataset and variant, averaged over 30 partitions. Success semantics differ between arms: A and B are SLSQP-certified, C is equality-feasible under the lenient rule. Showing 9 of 16 columns; the rest are in the source CSV.

dataset	variant	n subproblems	n success	n front real weighted	n front real support	total nfev	n real objective evals	seconds
bnp	A	66.000000	62.433300	13.800000	19.700000	3840.600000	0.000000	22.209200
bnp	B	66.000000	65.700000	45.933300	11.500000	2297.500000	0.000000	34.734400
bnp	C	66.000000	55.033300	33.600000	13.333300	105597.566700	416941.466700	1408.892900
porto	A	66.000000	37.966700	12.033300	3.733300	23230.233300	0.000000	83.130000
porto	B	66.000000	66.000000	54.966700	6.466700	4917.700000	0.000000	54.136100
porto	C	66.000000	64.533300	56.066700	12.200000	99642.533300	416846.400000	2502.923300
santander	A	66.000000	64.366700	12.800000	8.100000	5643.300000	0.000000	32.332500
santander	B	66.000000	62.633300	18.800000	11.133300	8003.033300	0.000000	32.896000
santander	C	66.000000	54.666700	33.366700	26.100000	101091.533300	412035.533300	2526.775700
uci_credit	A	66.000000	25.766700	15.700000	8.900000	33533.566700	0.000000	124.579800
uci_credit	B	66.000000	31.833300	30.533300	14.200000	28060.400000	0.000000	99.546500
uci_credit	C	66.000000	64.766700	62.000000	16.933300	101876.566700	425855.800000	306.357400

Success semantics are not on a common scale. NBI-A and NBI-B optimize smooth polynomials and a subproblem counts as successful only when SLSQP certifies optimality. NBI-C optimizes a piecewise-constant ROC-AUC, where such certification is generically unattainable, so a subproblem is accepted when its equality residual falls below 10^{-3}, under a reduced budget. The UCI credit gap between NBI-C and NBI-B is substantially a convergence effect of NBI-B, which certifies a median 29 of 66 subproblems there; retaining its unconverged iterates preserves the direction but reduces the mean gap from +0.277 to +0.108 hypervolume, and in the three partitions where NBI-B converges fully the gap is at most 0.01.

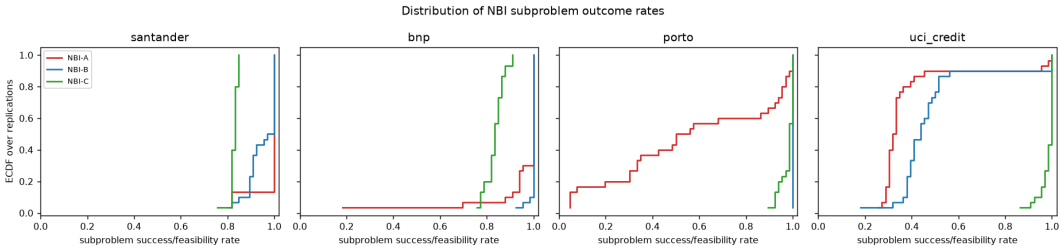


Figure S15: Empirical distribution of subproblem success rates per variant and dataset.

S8 Empirical reference convergence

Table S15: Empirical reference construction diagnostics, averaged over 30 partitions. Showing 6 of 21 columns; the rest are in the source CSV.

dataset	n points	rounds	displaced fraction	front weighted size	front support size
bnp	117230.666700	1.166700	0.020300	122.100000	38.533300
porto	137230.666700	1.366700	0.027200	102.433300	15.100000
santander	120564.000000	1.200000	0.027200	143.900000	42.333300
uci_credit	140564.000000	1.400000	0.036900	520.433300	91.433300

The reference has two layers. The *sampled core* is built from Dirichlet samples, the simplex lattice, all vertices and edges, an ε-constraint sweep, the 66 design runs and the single-objective references; it contains the search output of none of the five compared methods. The *final augmented reference* is the non-dominated union of that core with every candidate set. Because the candidate sets contribute a substantial share of its points, in proportions that differ by dataset and arm (NSGA-II, for example, supplies 8–29% of the points of the augmented union that includes it, averaged over four probed replications per dataset), absolute indicator values computed against the augmented reference are partly self-graded, which is why the primary comparisons are also scored against the sample core (selected contrasts in Section S5) and the NSGA-II comparison against both references (Section S13). On the weighted-cost front, 118 of the 120 replications met the 5% displacement tolerance and the other two ended at most 6% displaced (`tables/reference_diagnostics.csv`); the benchmark's check did not cover the support-cost front, and the independent check sample was not stored.

Support-cost convergence (post hoc). Table S16 computes, from the stored reference samples and with no composition evaluated, the same two diagnostics for both fronts. In place of the unstored independent sample, the last 20,000 points of the first-round random part, independent draws from the same shuffled mixture, check the front of the structured blocks plus the first 80,000 random points, a smaller reference than the one scored. As a calibration, this check passes the weighted-cost front in 26, 27, 22 and 9 of 30 partitions on Santander, BNP Paribas, Porto Seguro and UCI credit, against 26, 25, 21 and 18 for the benchmark's own check at its first round, before

the extension rounds that took it to 29–30 of 30: comparable on three datasets and stricter on UCI credit. On the held-out reading the support-cost front is less converged: it met the 5% tolerance in 7, 10, 12 and 6 of 30 partitions, with median displaced shares of 6.6–8.0% against 1.6–6.0% for the weighted front and a worst case of 45% (5 of 11 points) on one Porto Seguro partition. The prefix diagnostic on the random part alone is not decisive: neither front meets 5% at its last step in most partitions, the support-front medians are the higher, and with fronts this small a count at 5% carries little information. Support-cost fronts are small (median 14 to 87 points), so each displaced point is a large share. The reference was not enlarged and nothing was rescored; the main text qualifies the step-cost conclusions accordingly.

Table S16: Post hoc convergence check of both reference fronts from the stored samples (added in revision). Held-out block: share of the front of the structured blocks plus the first 80,000 random points that the remaining 20,000 random points displace; met: partitions at or below the 5% tolerance. Prefix: displaced share at the last prefix step (75% to 100% of the first-round random part). W: weighted-cost front; S: support-cost front; front: median stored front size; W recorded met: partitions meeting the benchmark's own weighted-cost check. Source: `scripts/pco213_support_reference_convergence.py`. Showing 11 of 15 columns; the rest are in the source CSV.

dataset	W met	S met	W median	S median	S max	W prefix	S prefix	W front	S front	W recorded met
santander	26	7	0.015700	0.078900	0.222200	0.111800	0.131900	135.000000	40.000000	29
bnp	27	10	0.019900	0.065600	0.184200	0.099300	0.160300	119.500000	39.000000	30
porto	22	12	0.024800	0.066700	0.454500	0.135800	0.154800	93.500000	14.000000	29
uci_credit	9	6	0.060000	0.080300	0.197700	0.103200	0.133300	481.000000	87.000000	30

S9 Cost definition and the accuracy–calibration conflict

Table S17: Cost-definition sensitivity in full. Showing 8 of 15 columns; the rest are in the source CSV.

dataset	endpoint	n	best set differs count	best set differs frac	wilson lo	wilson hi	median rank spearman weighted vs support
santander	hv_ratio	30	10.000000	0.333300	0.192300	0.512200	0.828600
santander	igd_plus	30	21.000000	0.700000	0.521200	0.833400	0.685700
santander	nbi_C_support_hv_ratio	30	–	–	–	–	–
bnp	hv_ratio	30	24.000000	0.800000	0.626900	0.904900	0.885700
bnp	igd_plus	30	20.000000	0.666700	0.487800	0.807700	0.771400
bnp	nbi_C_support_hv_ratio	30	–	–	–	–	–
porto	hv_ratio	30	8.000000	0.266700	0.141800	0.444500	0.942900
porto	igd_plus	30	2.000000	0.066700	0.018500	0.213200	1.000000
porto	nbi_C_support_hv_ratio	30	–	–	–	–	–
uci_credit	hv_ratio	30	20.000000	0.666700	0.487800	0.807700	0.657100
uci_credit	igd_plus	30	19.000000	0.633300	0.455100	0.781300	0.600000
uci_credit	nbi_C_support_hv_ratio	30	–	–	–	–	–

Table S18: AUC-versus-log-loss conflict in full, including weight-space distances, support Jaccard and both cost differences.

dataset	quantity	n	mean	median	sd	ci95 lo	ci95 hi	min	max
santander	delta_auc_directauc_minus_slsqp	30	0.000700	0.000700	0.000100	0.000700	0.000700	0.000500	0.000800
santander	delta_logloss_directauc_minus_slsqp	30	0.003200	0.003200	0.000500	0.003000	0.003400	0.002100	0.004600
santander	holdout_delta_auc	30	0.002000	0.002000	0.000300	0.001900	0.002100	0.001400	0.002600
santander	holdout_delta_logloss	30	0.001900	0.001900	0.000600	0.001600	0.002100	0.000900	0.003400
santander	weight_l1_distance	30	0.350400	0.349200	0.021900	0.342800	0.358300	0.304700	0.410800
santander	support_jaccard	30	0.504400	0.500000	0.084800	0.475600	0.534400	0.400000	0.666700
santander	delta_cost_weighted	30	2.861100	1.495000	3.618600	1.718100	4.252300	0.272800	15.829100
santander	delta_cost_support	30	183.384600	253.958400	111.917100	142.192700	221.033000	3.625600	257.801800
santander	ensembling_gain_auc_slsqp_minus_best_single	30	0.003700	0.003700	0.000200	0.003600	0.003700	0.003100	0.004200
bnp	delta_auc_directauc_minus_slsqp	30	0.000800	0.000800	0.000100	0.000800	0.000800	0.000600	0.001000
bnp	delta_logloss_directauc_minus_slsqp	30	0.000900	0.000900	0.000100	0.000900	0.000900	0.000800	0.001100
bnp	holdout_delta_auc	30	0.000300	0.000400	0.000400	0.000200	0.000400	-0.000400	0.001000
bnp	holdout_delta_logloss	30	0.001300	0.001400	0.000400	0.001200	0.001400	0.000500	0.002000
bnp	weight_l1_distance	30	0.376900	0.375600	0.019000	0.370500	0.383800	0.340900	0.437900
bnp	support_jaccard	30	0.758300	0.750000	0.045600	0.750000	0.775000	0.750000	1.000000
bnp	delta_cost_weighted	30	0.538600	0.532000	0.043900	0.523800	0.554200	0.428400	0.658600
bnp	delta_cost_support	30	-3.571500	-3.693700	0.676100	-3.706900	-3.317100	-3.780300	0.000000
bnp	ensembling_gain_auc_slsqp_minus_best_single	30	0.005600	0.005600	0.000400	0.005500	0.005800	0.004900	0.006400
porto	delta_auc_directauc_minus_slsqp	30	0.000100	0.000000	0.000000	0.000000	0.000100	0.000000	0.000200
porto	delta_logloss_directauc_minus_slsqp	30	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
porto	holdout_delta_auc	30	-0.000100	-0.000000	0.000300	-0.000200	0.000000	-0.001000	0.000400
porto	holdout_delta_logloss	30	0.000000	0.000000	0.000000	0.000000	0.000000	-0.000000	0.000100
porto	weight_l1_distance	30	0.058400	0.053600	0.021800	0.050800	0.066300	0.023200	0.107100
porto	support_jaccard	30	0.973300	1.000000	0.069100	0.946700	0.993300	0.800000	1.000000
porto	delta_cost_weighted	30	2.692900	2.798800	2.164800	1.925100	3.436500	-2.338100	7.188000
porto	delta_cost_support	30	-0.105800	0.000000	0.578200	-0.317400	0.104900	-1.595500	1.579900
porto	ensembling_gain_auc_slsqp_minus_best_single	30	0.006000	0.005700	0.001100	0.005600	0.006300	0.003700	0.008400
uci_credit	delta_auc_directauc_minus_slsqp	30	0.000300	0.000300	0.000100	0.000300	0.000400	0.000200	0.000600
uci_credit	delta_logloss_directauc_minus_slsqp	30	0.000500	0.000500	0.000200	0.000400	0.000600	0.000200	0.000900
uci_credit	holdout_delta_auc	30	0.000300	0.000500	0.000500	0.000200	0.000500	-0.000600	0.001100
uci_credit	holdout_delta_logloss	30	0.000400	0.000300	0.000400	0.000200	0.000500	-0.000400	0.001500
uci_credit	weight_l1_distance	30	0.200900	0.203900	0.043800	0.185700	0.216200	0.097600	0.295600
uci_credit	support_jaccard	30	0.866700	0.800000	0.138500	0.816700	0.915000	0.600000	1.000000
uci_credit	delta_cost_weighted	30	0.884600	0.802900	0.391200	0.752700	1.023900	0.169200	1.742700
uci_credit	delta_cost_support	30	0.302200	0.255700	0.393300	0.165400	0.442500	-0.565700	1.081000
uci_credit	ensembling_gain_auc_slsqp_minus_best_single	30	0.000600	0.000600	0.000300	0.000500	0.000800	0.000200	0.001400

The accuracy–calibration conflict in weight space, and the Santander cost artifact. The ROC-AUC advantage of the direct optimum over the SLSQP log-loss optimum has bootstrap interval half-widths below 0.00004 on every dataset. The two optima nonetheless differ in weight space: L_1 distances 0.35, 0.38, 0.06 and 0.20 (Santander, BNP Paribas, Porto Seguro, UCI credit), the Porto Seguro pair practically coinciding, with support Jaccard 0.50 to 0.97. The apparent +254 ms per 1,000 rows Santander deployment-cost gap between the two optima is an activity-threshold artifact: the ROC-AUC optimum retains a k-nearest-neighbour weight just above 10^{-3} that is AUC-inert (zeroing it changes out-of-fold ROC-AUC by 3×10^{-6} on average), and an AUC-equivalent blend without k-nearest neighbours costs about +4 ms per 1,000 rows. No deployment-cost gap between the two optima is claimed. Under the two cost definitions the hypervolume-best method changes in 10 of 30 partitions on Santander, 24 of 30 on BNP Paribas, 8 of 30 on Porto Seguro and 20 of 30 on UCI credit.

The BNP Paribas support-cost normalization artifact. NBI-C's support-cost hypervolume appears bimodal, but this is a per-replication normalization effect, not a regime. The cost bound of the normalization box is approximately 17 ms per 1,000 rows in 13 partitions, where no k-nearest-neighbour-bearing point survives on the support reference front, and 160–167 in the other 17, a tenfold rescaling of one axis. NBI-C's fronts are the same in both groups. Under a common bound the split disappears. The robust statement is qualitative: NBI sets optimized under the linear cost omit the cheapest supports that the raw design contains.

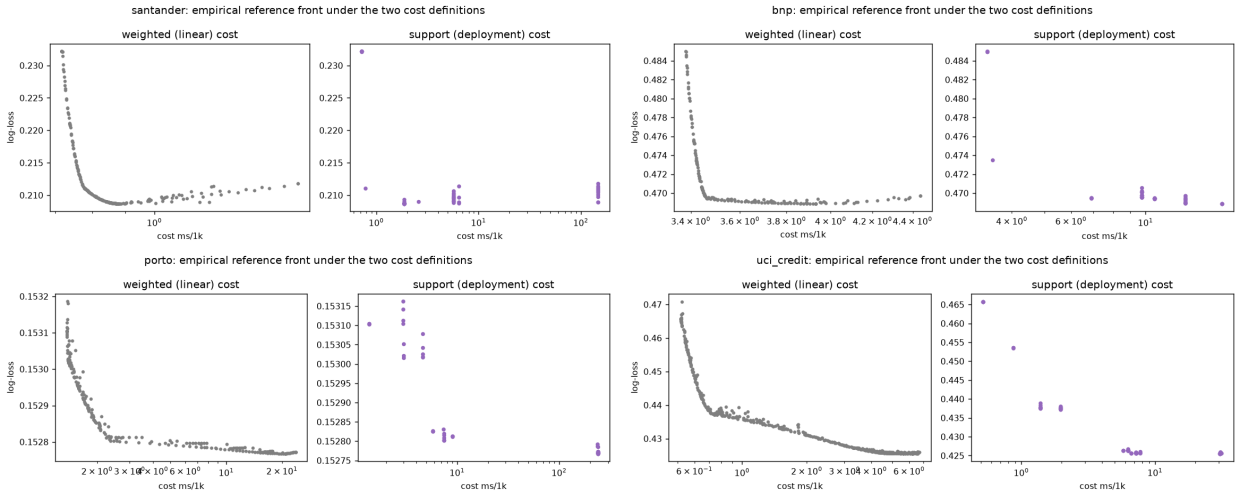


Figure S16: Weighted versus support cost, all four datasets.

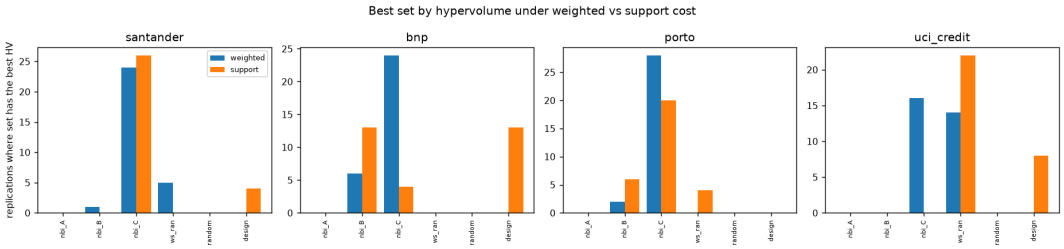


Figure S17: Set rankings under the two cost definitions.

S10 Holdout transfer

Table S19: Holdout transfer for every set under both cost definitions. Showing 12 of 16 columns; the rest are in the source CSV.

cost	dataset	set	metric	n	oof mean	holdout mean	delta mean	delta ci95 lo	delta ci95 hi	frac abs delta gt 0.005	wins
weighted	santander	nbi_A	roc_auc	30	0.891100	0.887200	-0.003900	-0.005100	-0.002600	0.366700	–
weighted	santander	nbi_A	log_loss	30	0.212700	0.215100	0.002500	0.001600	0.003300	0.100000	–
weighted	santander	nbi_B	roc_auc	30	0.883000	0.881100	-0.001800	-0.003100	-0.000600	0.200000	–
weighted	santander	nbi_B	log_loss	30	0.216200	0.217500	0.001300	0.000500	0.002100	0.033300	–
weighted	santander	nbi_C	roc_auc	30	0.886700	0.884800	-0.001800	-0.003000	-0.000700	0.166700	–
weighted	santander	nbi_C	log_loss	30	0.213400	0.214600	0.001200	0.000400	0.001900	0.066700	–
weighted	santander	ws_random_scalarization	roc_auc	30	0.888700	0.884100	-0.004600	-0.005900	-0.003400	0.500000	–
weighted	santander	ws_random_scalarization	log_loss	30	0.210800	0.214400	0.003600	0.002700	0.004500	0.233300	–
weighted	santander	ALL(knee picks)	ranking_agreement_auc	30	–	–	1.000000	0.886500	1.000000	–	30.000000
weighted	bnp	nbi_A	roc_auc	30	0.746900	0.748600	0.001700	0.000200	0.003300	0.333300	–
weighted	bnp	nbi_A	log_loss	30	0.479800	0.478600	-0.001200	-0.002200	-0.000300	0.100000	–
weighted	bnp	nbi_B	roc_auc	30	0.750500	0.752200	0.001700	0.000200	0.003200	0.300000	–
weighted	bnp	nbi_B	log_loss	30	0.472100	0.470900	-0.001100	-0.002100	-0.000200	0.133300	–
weighted	bnp	nbi_C	roc_auc	30	0.750700	0.752300	0.001600	0.000100	0.003100	0.266700	–
weighted	bnp	nbi_C	log_loss	30	0.472200	0.471200	-0.001000	-0.001900	-0.000100	0.100000	–
weighted	bnp	ws_random_scalarization	roc_auc	30	0.743800	0.747100	0.003300	0.001700	0.005000	0.400000	–
weighted	bnp	ws_random_scalarization	log_loss	30	0.480500	0.477800	-0.002800	-0.004000	-0.001600	0.233300	–
weighted	bnp	ALL(knee picks)	ranking_agreement_auc	30	–	–	0.500000	0.331500	0.668500	–	15.000000
weighted	porto	nbi_A	roc_auc	30	0.627500	0.629200	0.001700	-0.001900	0.005500	0.666700	–
weighted	porto	nbi_A	log_loss	30	0.152800	0.152700	-0.000100	-0.000300	0.000100	0.000000	–
weighted	porto	nbi_B	roc_auc	30	0.628900	0.630700	0.001800	-0.001800	0.005500	0.633300	–
weighted	porto	nbi_B	log_loss	30	0.152600	0.152500	-0.000100	-0.000300	0.000100	0.000000	–
weighted	porto	nbi_C	roc_auc	30	0.630500	0.632400	0.001900	-0.001600	0.005600	0.533300	–
weighted	porto	nbi_C	log_loss	30	0.152500	0.152400	-0.000100	-0.000300	0.000100	0.000000	–
weighted	porto	ws_random_scalarization	roc_auc	30	0.627000	0.629000	0.002000	-0.001600	0.005800	0.600000	–
weighted	porto	ws_random_scalarization	log_loss	30	0.154000	0.153900	-0.000100	-0.000400	0.000200	0.000000	–
weighted	porto	ALL(knee picks)	ranking_agreement_auc	30	–	–	0.766700	0.590700	0.882100	–	23.000000
weighted	uci_credit	nbi_A	roc_auc	30	0.760700	0.760700	0.000000	-0.003100	0.003100	0.600000	–
weighted	uci_credit	nbi_A	log_loss	30	0.441300	0.441300	0.000000	-0.002000	0.002100	0.433300	–
weighted	uci_credit	nbi_B	roc_auc	30	0.775400	0.775700	0.000300	-0.002500	0.003100	0.600000	–
weighted	uci_credit	nbi_B	log_loss	30	0.432800	0.432800	-0.000000	-0.001900	0.002000	0.333300	–
weighted	uci_credit	nbi_C	roc_auc	30	0.768100	0.769500	0.001400	-0.001300	0.004200	0.600000	–
weighted	uci_credit	nbi_C	log_loss	30	0.437200	0.436200	-0.000900	-0.003000	0.001000	0.433300	–
weighted	uci_credit	ws_random_scalarization	roc_auc	30	0.771300	0.772300	0.000900	-0.001800	0.003700	0.566700	–
weighted	uci_credit	ws_random_scalarization	log_loss	30	0.435400	0.434700	-0.000800	-0.002900	0.001300	0.466700	–
weighted	uci_credit	ALL(knee picks)	ranking_agreement_auc	30	–	–	1.000000	0.886500	1.000000	–	30.000000
support	santander	nbi_A	roc_auc	30	0.890400	0.885500	-0.004900	-0.006000	-0.003800	0.533300	–
support	santander	nbi_A	log_loss	30	0.211200	0.214700	0.003500	0.002700	0.004300	0.166700	–
support	santander	nbi_B	roc_auc	30	0.890900	0.887300	-0.003600	-0.004700	-0.002400	0.300000	–
support	santander	nbi_B	log_loss	30	0.209900	0.212300	0.002400	0.001600	0.003100	0.066700	–
support	santander	nbi_C	roc_auc	30	0.892100	0.889300	-0.002700	-0.003800	-0.001700	0.233300	–
support	santander	nbi_C	log_loss	30	0.209400	0.211000	0.001600	0.000800	0.002300	0.066700	–
support	santander	ws_random_scalarization	roc_auc	30	0.891100	0.886700	-0.004300	-0.005500	-0.003200	0.400000	–
support	santander	ws_random_scalarization	log_loss	30	0.209400	0.212600	0.003200	0.002300	0.004000	0.266700	–
support	santander	ALL(knee picks)	ranking_agreement_auc	30	–	–	0.933300	0.786800	0.981500	–	28.000000
support	bnp	nbi_A	roc_auc	30	0.746400	0.751000	0.004600	0.002900	0.006300	0.500000	–
support	bnp	nbi_A	log_loss	30	0.481800	0.478100	-0.003700	-0.004900	-0.002500	0.333300	–
support	bnp	nbi_B	roc_auc	30	0.752800	0.755500	0.002800	0.001200	0.004300	0.333300	–
support	bnp	nbi_B	log_loss	30	0.469700	0.467800	-0.001900	-0.002900	-0.000800	0.166700	–
support	bnp	nbi_C	roc_auc	30	0.752000	0.754500	0.002600	0.001000	0.004000	0.400000	–
support	bnp	nbi_C	log_loss	30	0.470700	0.469000	-0.001700	-0.002700	-0.000700	0.166700	–
support	bnp	ws_random_scalarization	roc_auc	30	0.745400	0.749200	0.003800	0.002100	0.005600	0.433300	–
support	bnp	ws_random_scalarization	log_loss	30	0.479500	0.476500	-0.003100	-0.004300	-0.001900	0.266700	–

Continued on next page

cost	dataset	set	metric	n	oof mean	holdout mean	delta mean	delta ci95 lo	delta ci95 hi	frac abs delta gt 0.005	wins
support	bnp	ALL(knee picks)	ranking_agreement_auc	30	–	–	0.933300	0.786800	0.981500	–	28.000000
support	porto	nbi_A	roc_auc	30	0.627200	0.629300	0.002100	-0.001500	0.005900	0.666700	–
support	porto	nbi_A	log_loss	30	0.152700	0.152600	-0.000100	-0.000400	0.000100	0.000000	–
support	porto	nbi_B	roc_auc	30	0.630200	0.631900	0.001700	-0.001900	0.005400	0.533300	–
support	porto	nbi_B	log_loss	30	0.152500	0.152400	-0.000100	-0.000300	0.000100	0.000000	–
support	porto	nbi_C	roc_auc	30	0.630600	0.632500	0.001900	-0.001600	0.005600	0.533300	–
support	porto	nbi_C	log_loss	30	0.152500	0.152400	-0.000100	-0.000300	0.000100	0.000000	–
support	porto	ws_random_scalarization	roc_auc	30	0.627400	0.629400	0.002100	-0.001600	0.005800	0.633300	–
support	porto	ws_random_scalarization	log_loss	30	0.153900	0.153800	-0.000100	-0.000400	0.000100	0.000000	–
support	porto	ALL(knee picks)	ranking_agreement_auc	30	–	–	0.566700	0.392000	0.726200	–	17.000000
support	uci_credit	nbi_A	roc_auc	30	0.767900	0.768200	0.000300	-0.002900	0.003500	0.633300	–
support	uci_credit	nbi_A	log_loss	30	0.436800	0.436700	-0.000100	-0.002300	0.002100	0.466700	–
support	uci_credit	nbi_B	roc_auc	30	0.783200	0.783500	0.000300	-0.002700	0.003200	0.633300	–
support	uci_credit	nbi_B	log_loss	30	0.427300	0.427200	-0.000100	-0.002200	0.002000	0.400000	–
support	uci_credit	nbi_C	roc_auc	30	0.783400	0.783600	0.000200	-0.002700	0.003100	0.633300	–
support	uci_credit	nbi_C	log_loss	30	0.427100	0.427100	-0.000100	-0.002100	0.002000	0.433300	–
support	uci_credit	ws_random_scalarization	roc_auc	30	0.776800	0.777000	0.000200	-0.002800	0.003300	0.700000	–
support	uci_credit	ws_random_scalarization	log_loss	30	0.431800	0.431600	-0.000200	-0.002300	0.001900	0.433300	–
support	uci_credit	ALL(knee picks)	ranking_agreement_auc	30	–	–	0.533300	0.361400	0.697700	–	16.000000

Ranking agreement by batch. The out-of-fold-best of the four knee-selected picks is also holdout-best in 30/30 partitions on Santander, 15/30 on BNP Paribas, 23/30 on Porto Seguro and 30/30 on UCI credit; restricted to the twenty partitions added in the extension the counts are 20/20, 10/20, 15/20 and 20/20, reproducing the pattern exactly.

Knee-selected solutions compared across arms. Pairing the knee-selected solution of two arms by partition under the weighted cost, the two NBI contrasts that are Holm-significant on both primary endpoints have worse single-objective holdout accuracy at the selected point: Santander NBI-B minus NBI-A is −0.0047 holdout ROC-AUC, better in 15 of 30 partitions, with holdout log-loss worse in 16 of 30; UCI credit NBI-C minus NBI-B is −0.0119, better in 6 of 30, with log-loss worse in 24 of 30. Both selected points are also materially cheaper, by 0.97 and 2.84 ms per 1,000 rows. The two contrasts in which the selected point does improve are BNP Paribas NBI-B minus NBI-A (+0.0029, better in 28 of 30) and Santander NBI-C minus NBI-B (+0.0016, better in 26 of 30), the contrasts with the smaller indicator effects.

How much there is to win at all. The gain of the log-loss optimum over the best single model on the untouched holdout is +0.0076, +0.0038, +0.0053 and +0.0005 ROC-AUC on Santander, BNP Paribas, Porto Seguro and UCI credit, better than the best single model in 30, 30, 29 and 23 of 30 partitions. On UCI credit the out-of-fold gain is +0.0007 and the Nadeau–Bengio-corrected p is 0.24: on that dataset no weighting method in this study separates from the best single model on held-out data.

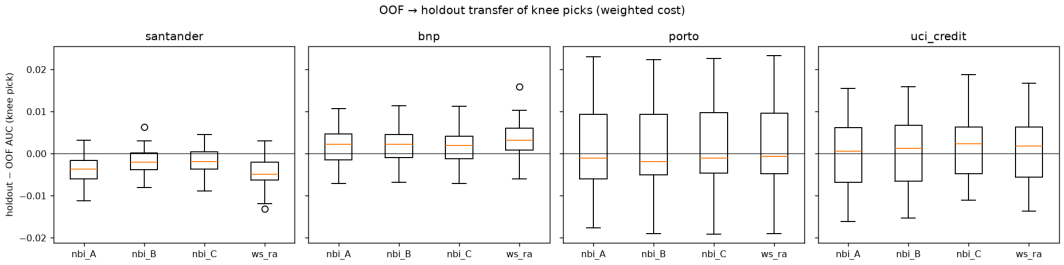


Figure S18: Holdout minus out-of-fold differences for every knee-selected solution.

S11 From ten to thirty replications

Table S20: Complete $R = 10$ versus $R = 30$ comparison across all sixteen metric families. First 90 of 352 rows; the complete table is the source CSV. Showing 12 of 16 columns; the rest are in the source CSV.

metric	dataset	response	set	cost	term	r10 estimate	r30 estimate	abs change	rel change	r30 ci95 lo	r30 ci95 hi
scheffe_r2_external	santander	roc_auc	–	–	–	-0.275500	-0.371200	-0.095700	-0.347400	-0.558500	-0.199000
reliability_gate_pass	santander	roc_auc	–	–	–	0.000000	0.000000	0.000000	–	0.000000	0.000000
scheffe_r2_external	santander	log_loss	–	–	–	0.973100	0.970000	-0.003100	-0.003200	0.967300	0.972500
reliability_gate_pass	santander	log_loss	–	–	–	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
scheffe_r2_external	bnp	roc_auc	–	–	–	-0.008700	0.112500	0.121300	13.874600	-0.034500	0.244100
reliability_gate_pass	bnp	roc_auc	–	–	–	0.200000	0.166700	-0.033300	-0.166700	0.033300	0.300000
scheffe_r2_external	bnp	log_loss	–	–	–	0.086800	-0.104200	-0.191000	-2.200500	-0.245900	0.041800
reliability_gate_pass	bnp	log_loss	–	–	–	0.200000	0.100000	-0.100000	-0.500000	0.000000	0.200000
scheffe_r2_external	porto	roc_auc	–	–	–	0.519600	0.530700	0.011000	0.021200	0.482200	0.580500
reliability_gate_pass	porto	roc_auc	–	–	–	0.600000	0.533300	-0.066700	-0.111100	0.366700	0.700000
scheffe_r2_external	porto	log_loss	–	–	–	0.609200	0.535700	-0.073500	-0.120700	0.492200	0.579800
reliability_gate_pass	porto	log_loss	–	–	–	0.500000	0.400000	-0.100000	-0.200000	0.233300	0.566700
scheffe_r2_external	uci_credit	roc_auc	–	–	–	0.992800	0.990700	-0.002000	-0.002000	0.989200	0.992100
reliability_gate_pass	uci_credit	roc_auc	–	–	–	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
scheffe_r2_external	uci_credit	log_loss	–	–	–	0.993800	0.991900	-0.001900	-0.001900	0.990500	0.993200
reliability_gate_pass	uci_credit	log_loss	–	–	–	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_linear	santander	roc_auc	–	–	gnb	0.886100	0.886300	0.000200	0.000200	0.886100	0.886500
beta_linear	santander	roc_auc	–	–	knn	0.757100	0.756500	-0.000600	-0.000800	0.755900	0.757200
beta_linear	santander	roc_auc	–	–	lr	0.857800	0.858000	0.000200	0.000200	0.857700	0.858300
beta_linear	santander	roc_auc	–	–	rf	0.849000	0.849300	0.000300	0.000300	0.848900	0.849700
beta_linear	santander	roc_auc	–	–	xgb	0.880000	0.880300	0.000300	0.000300	0.880000	0.880600
beta_interaction_top3	santander	roc_auc	–	–	gnb*knn	0.242300	0.243200	0.000900	0.003600	0.242300	0.244200
beta_interaction_sign_pos_freq	santander	roc_auc	–	–	gnb*knn	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_interaction_top3	santander	roc_auc	–	–	knn*xgb	0.231100	0.232000	0.000900	0.003900	0.231100	0.232900
beta_interaction_sign_pos_freq	santander	roc_auc	–	–	knn*xgb	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_interaction_top3	santander	roc_auc	–	–	knn*rf	0.198700	0.199300	0.000600	0.002900	0.198500	0.200000
beta_interaction_sign_pos_freq	santander	roc_auc	–	–	knn*rf	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_linear	santander	log_loss	–	–	gnb	0.211600	0.211400	-0.000200	-0.000700	0.211300	0.211600
beta_linear	santander	log_loss	–	–	knn	0.340200	0.340500	0.000300	0.000900	0.340300	0.340800
beta_linear	santander	log_loss	–	–	lr	0.232300	0.232200	-0.000100	-0.000500	0.232100	0.232400
beta_linear	santander	log_loss	–	–	rf	0.275100	0.275000	-0.000000	-0.000100	0.274900	0.275100
beta_linear	santander	log_loss	–	–	xgb	0.220100	0.220000	-0.000100	-0.000600	0.219800	0.220200
beta_interaction_top3	santander	log_loss	–	–	gnb*knn	-0.173500	-0.174000	-0.000500	-0.002900	-0.174400	-0.173600
beta_interaction_sign_pos_freq	santander	log_loss	–	–	gnb*knn	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_interaction_top3	santander	log_loss	–	–	lr*knn	-0.147200	-0.147600	-0.000400	-0.002900	-0.148000	-0.147200
beta_interaction_sign_pos_freq	santander	log_loss	–	–	lr*knn	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_interaction_top3	santander	log_loss	–	–	knn*xgb	-0.136300	-0.136700	-0.000500	-0.003500	-0.137200	-0.136300
beta_interaction_sign_pos_freq	santander	log_loss	–	–	knn*xgb	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_linear	bnp	roc_auc	–	–	gnb	0.633900	0.637200	0.003300	0.005200	0.634200	0.640000
beta_linear	bnp	roc_auc	–	–	knn	0.682300	0.682000	-0.000300	-0.000400	0.681500	0.682500
beta_linear	bnp	roc_auc	–	–	lr	0.734100	0.734300	0.000100	0.000200	0.734000	0.734600
beta_linear	bnp	roc_auc	–	–	rf	0.742900	0.743000	0.000000	0.000000	0.742600	0.743300
beta_linear	bnp	roc_auc	–	–	xgb	0.746900	0.747100	0.000300	0.000400	0.746600	0.747700
beta_interaction_top3	bnp	roc_auc	–	–	gnb*xgb	0.157200	0.147800	-0.009400	-0.059900	0.139100	0.157000
beta_interaction_sign_pos_freq	bnp	roc_auc	–	–	gnb*xgb	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_interaction_top3	bnp	roc_auc	–	–	gnb*rf	0.147200	0.138600	-0.008600	-0.058600	0.130400	0.147300
beta_interaction_sign_pos_freq	bnp	roc_auc	–	–	gnb*rf	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_interaction_top3	bnp	roc_auc	–	–	lr*gnb	0.141300	0.132900	-0.008300	-0.058900	0.124900	0.141500
beta_interaction_sign_pos_freq	bnp	roc_auc	–	–	lr*gnb	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_linear	bnp	log_loss	–	–	gnb	3.104300	3.035500	-0.068800	-0.022200	2.978200	3.095200
beta_linear	bnp	log_loss	–	–	knn	0.470100	0.471300	0.001300	0.002700	0.470300	0.472300

Continued on next page

metric	dataset	response	set	cost	term	r10 estimate	r30 estimate	abs change	rel change	r30 ci95 lo	r30 ci95 hi
beta_linear	bnp	log_loss	–	–	lr	0.441500	0.442600	0.001100	0.002500	0.441600	0.443500
beta_linear	bnp	log_loss	–	–	rf	0.439900	0.441000	0.001100	0.002500	0.440100	0.441900
beta_linear	bnp	log_loss	–	–	xgb	0.431100	0.432200	0.001100	0.002600	0.431200	0.433100
beta_interaction_top3	bnp	log_loss	–	–	gnb*xgb	-4.023200	-3.921400	0.101800	0.025300	-4.010300	-3.836000
beta_interaction_sign_pos_freq	bnp	log_loss	–	–	gnb*xgb	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_interaction_top3	bnp	log_loss	–	–	gnb*rf	-3.992900	-3.891700	0.101200	0.025300	-3.979700	-3.807200
beta_interaction_sign_pos_freq	bnp	log_loss	–	–	gnb*rf	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_interaction_top3	bnp	log_loss	–	–	gnb*knn	-3.987000	-3.888100	0.098900	0.024800	-3.974600	-3.805000
beta_interaction_sign_pos_freq	bnp	log_loss	–	–	gnb*knn	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_linear	porto	roc_auc	–	–	gnb	0.594700	0.593900	-0.000800	-0.001400	0.593100	0.594600
beta_linear	porto	roc_auc	–	–	knn	0.568700	0.568600	-0.000100	-0.000100	0.567600	0.569700
beta_linear	porto	roc_auc	–	–	lr	0.624700	0.625500	0.000700	0.001200	0.624500	0.626400
beta_linear	porto	roc_auc	–	–	rf	0.620000	0.621300	0.001300	0.002100	0.620300	0.622300
beta_linear	porto	roc_auc	–	–	xgb	0.591800	0.592800	0.001000	0.001700	0.591600	0.594000
beta_interaction_top3	porto	roc_auc	–	–	gnb*knn	0.071400	0.070700	-0.000700	-0.010200	0.067800	0.073500
beta_interaction_sign_pos_freq	porto	roc_auc	–	–	gnb*knn	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_interaction_top3	porto	roc_auc	–	–	knn*xgb	0.067600	0.067900	0.000300	0.004600	0.066700	0.069100
beta_interaction_sign_pos_freq	porto	roc_auc	–	–	knn*xgb	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_interaction_top3	porto	roc_auc	–	–	lr*knn	0.062400	0.062800	0.000400	0.006300	0.061900	0.063800
beta_interaction_sign_pos_freq	porto	roc_auc	–	–	lr*knn	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_linear	porto	log_loss	–	–	gnb	1.551800	1.532100	-0.019700	-0.012700	1.472100	1.590200
beta_linear	porto	log_loss	–	–	knn	0.141500	0.141700	0.000200	0.001600	0.140800	0.142500
beta_linear	porto	log_loss	–	–	lr	0.134200	0.134300	0.000200	0.001200	0.133500	0.135100
beta_linear	porto	log_loss	–	–	rf	0.134600	0.134700	0.000100	0.000900	0.133800	0.135500
beta_linear	porto	log_loss	–	–	xgb	0.139700	0.139800	0.000100	0.001000	0.138900	0.140600
beta_interaction_top3	porto	log_loss	–	–	gnb*knn	-1.925600	-1.901500	0.024200	0.012600	-1.981000	-1.821500
beta_interaction_sign_pos_freq	porto	log_loss	–	–	gnb*knn	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_interaction_top3	porto	log_loss	–	–	gnb*xgb	-1.917600	-1.893200	0.024400	0.012700	-1.972600	-1.813200
beta_interaction_sign_pos_freq	porto	log_loss	–	–	gnb*xgb	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_interaction_top3	porto	log_loss	–	–	gnb*rf	-1.910100	-1.886300	0.023800	0.012500	-1.965100	-1.807100
beta_interaction_sign_pos_freq	porto	log_loss	–	–	gnb*rf	0.000000	0.000000	0.000000	–	0.000000	0.000000
beta_linear	uci_credit	roc_auc	–	–	gnb	0.664200	0.663300	-0.001000	-0.001500	0.660800	0.665900
beta_linear	uci_credit	roc_auc	–	–	knn	0.757100	0.757500	0.000400	0.000500	0.756700	0.758200
beta_linear	uci_credit	roc_auc	–	–	lr	0.722100	0.722400	0.000300	0.000500	0.721800	0.723000
beta_linear	uci_credit	roc_auc	–	–	rf	0.782900	0.782800	-0.000100	-0.000100	0.782200	0.783500
beta_linear	uci_credit	roc_auc	–	–	xgb	0.763300	0.763000	-0.000300	-0.000400	0.762200	0.763700
beta_interaction_top3	uci_credit	roc_auc	–	–	gnb*xgb	0.128700	0.130100	0.001400	0.010900	0.127200	0.132800
beta_interaction_sign_pos_freq	uci_credit	roc_auc	–	–	gnb*xgb	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000
beta_interaction_top3	uci_credit	roc_auc	–	–	gnb*rf	0.101400	0.102800	0.001400	0.013800	0.100300	0.105300

What tripling the replications changed. Three findings changed status between $R = 10$ and $R = 30$. The Porto Seguro and UCI credit anchor effect went from an undecidable 6/10 and 5/10 with one tie to 24/30 on hypervolume (23–24/30 on IGD^+) with bootstrap intervals excluding zero, consistent in direction but not statistically established (Holm-adjusted Nadeau–Bengio $p = 0.79$ on Porto Seguro and 0.45 on UCI credit, on both endpoints); the BNP Paribas anchor collapse became a quantified rate of 7/30 with an identified trigger, selection of the quadratic ROC-AUC order (Section S6); and two cost readings, the BNP Paribas support-cost bimodality and the Santander deployment-cost gap, were shown to be artifacts (Section S9), the first of them noted but not resolved in the ten-partition analysis, the second not recorded there.

Against the twenty new partitions alone, rather than the nested $R = 30$ sample, only 6 of 12 hypervolume and 5 of 12 IGD^+ cells agree with the $R = 10$ estimate; the shift is clearest on Santander, whose IGD^+ is higher in the new partitions for all three NBI variants. The hypervolume anchor win rates are 6/10 against 18/20 on Porto Seguro (Fisher $p = 0.14$) and 5/10 against 19/20 on UCI credit ($p = 0.009$), both uncorrected. Analysed separately, the two batches give the same sign of the median paired effect in 28 of the 32 cells of the primary analysis, including every effect established at $R = 30$; the four exceptions, which are also the only cells whose sign differs between $R = 10$ and $R = 30$, are listed in Section 5.9 of the main text. These batch comparisons were not in the statistical analysis plan and are exploratory, and the difference between the batches is unexplained (run conditions in Section 7 of the main text); they show how far ten partitions can sit from the next twenty.

Provenance of the two batches (added in revision; post hoc). Execution date alone should not change deterministic objectives, so one bounded check of the existing records was made; no learner or optimizer was rerun. *Invariant:* the library source (`src/`) is identical for partitions 1–29, which ran under the solver-fixed code; the recorded environment block (Python, NumPy, pandas, SciPy, scikit-learn, statsmodels and XGBoost versions, platform, 14 cores) is identical in the two batch manifests; the base seed 20260904 and its per-replication rule are the same; the configuration differs only in the replication count and in two flags that compute nothing (`no_auto_reduce`, `dry_run`), and the automatic budget reduction never triggered; the dataset checksums agree. *Changed:* runner bookkeeping between the batches, which alters no computation; the execution order and timing (a pilot pass over partition 0, then blocks by dataset, against blocks by dataset in the extension); within the first batch, the 32 stages of partition 0 run under the pre-fix commit `c2e37c4` (Section S1); and the re-measured wall-clock costs, whose batch medians agree within about 4%. *Not recorded:* whether the working tree had uncommitted changes, a dependency lockfile, the linear-algebra backend and the versions of joblib and threadpoolctl, thread-related environment variables, power, thermal state and concurrent load, and checksums of the per-partition outputs. *Nondeterminism:* no partition was computed in both batches, so a direct determinism check across them is impossible; the one recomputation available, the NBI stage of UCI credit partition 0 rerun from its stored inputs between the batches, reproduced every stored candidate exactly. The records therefore neither explain the difference nor show nondeterminism, and it remains unexplained.

S12 Anchor-injection control

Table S21: Anchor-injection control: median over partitions of the fraction of the NBI-A to NBI-B hypervolume gap closed by adding the three real anchors to NBI-A's own candidate set, and the residual attributable to the relocated CHIM. The fraction closed is a median over the partitions in which NBI-B improves on NBI-A, the only ones where it is defined; the other columns are medians over all 30 partitions.

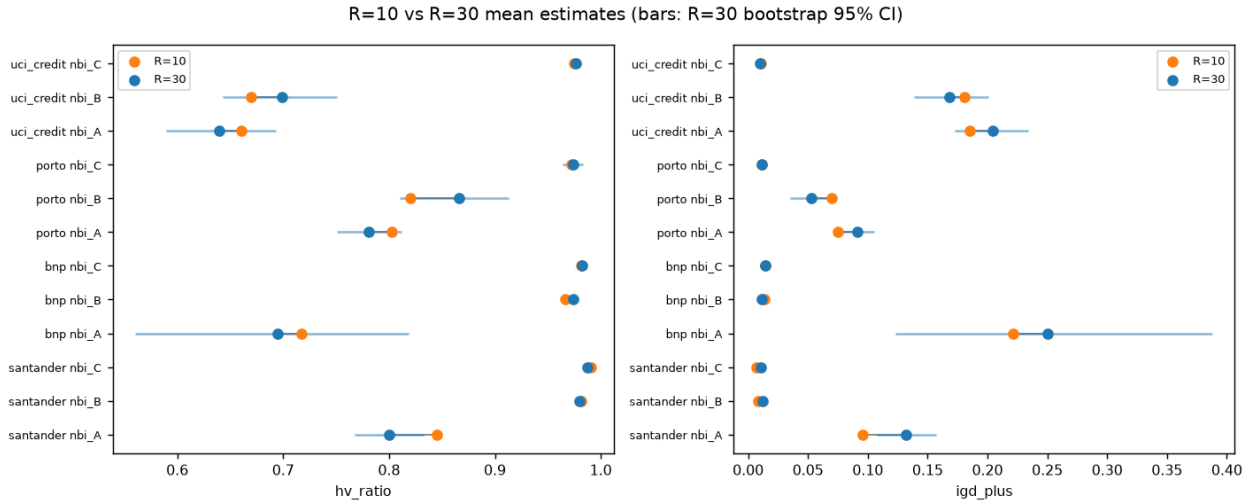


Figure S19: $R = 10$ against $R = 30$ across all metric families.

dataset	n	hv A	hv B	hv A plus anchors	fraction closed	residual chim
bnp	30	0.913000	0.971000	0.964000	0.898200	0.005100
porto	30	0.769000	0.913700	0.832100	0.485600	0.021500
santander	30	0.788700	0.980700	0.943900	0.797100	0.038400
uci_credit	30	0.621500	0.706200	0.623000	0.009300	0.090800

Because a vertex value of β returns its anchor directly, NBI-B's returned set contains the three real single-objective optima, which are also points of the empirical reference. Part of the NBI-B advantage is therefore the injection of those extreme solutions rather than the relocation of the CHIM. Rescoring NBI-A's own candidate set augmented with the same three anchors, and changing nothing else, separates the injection from the rest; the residual still bundles the relocated utopia point, normalization and CHIM, which no control here separates. Source: `scripts/pco213_anchor_injection_control.py` and `reports/pco213_postwork_benchmark/anchor_injection_control.csv`.

S13 External NSGA-II baseline

Table S22: NSGA-II paired effects against NBI-C, NBI-B and random scalarization, under both reference definitions and both cost definitions. First 70 of 96 rows; the complete table is the source CSV. Showing 15 of 31 columns; the rest are in the source CSV.

reference	cost	dataset	comparison	endpoint	n	mean	median	ci95 median lo	ci95 median hi	wins	ties	losses	rank biserial	p nb
augmented_union	support	santander	nsga2 vs nbi_C	igd_plus	30	-0.000300	-0.000300	-0.000700	0.000100	11	2	17	-0.277400	0.659800
augmented_union	support	santander	nsga2 vs nbi_C	hv_ratio	30	0.004300	0.002600	0.001200	0.004800	29	0	1	0.888200	0.231300
augmented_union	support	santander	nsga2 vs nbi_B	igd_plus	30	0.006700	0.006500	0.005900	0.007700	30	0	0	1.000000	0.000000
augmented_union	support	santander	nsga2 vs nbi_B	hv_ratio	30	0.009200	0.011100	0.005600	0.012200	29	0	1	0.982800	0.005900
augmented_union	support	santander	nsga2 vs ws_random_scalarization	igd_plus	30	0.008500	0.007900	0.007400	0.008600	30	0	0	1.000000	0.000800
augmented_union	support	santander	nsga2 vs ws_random_scalarization	hv_ratio	30	0.008200	0.010800	0.010400	0.011800	28	0	2	0.866700	0.472400
augmented_union	support	bnp	nsga2 vs nbi_C	igd_plus	30	0.030600	0.016900	0.003500	0.026400	28	0	2	0.982800	0.175800
augmented_union	support	bnp	nsga2 vs nbi_C	hv_ratio	30	0.116100	0.041100	0.018800	0.227300	30	0	0	1.000000	0.068700
augmented_union	support	bnp	nsga2 vs nbi_B	igd_plus	30	0.017700	0.009700	0.003400	0.016000	23	0	7	0.810800	0.182300
augmented_union	support	bnp	nsga2 vs nbi_B	hv_ratio	30	0.066800	0.016700	0.010300	0.056900	24	0	6	0.776300	0.176500
augmented_union	support	bnp	nsga2 vs ws_random_scalarization	igd_plus	30	0.394400	0.317200	0.220800	0.381500	29	0	1	0.995700	0.073800
augmented_union	support	bnp	nsga2 vs ws_random_scalarization	hv_ratio	30	0.409400	0.358300	0.267500	0.472200	30	0	0	1.000000	0.001600
augmented_union	support	porto	nsga2 vs nbi_C	igd_plus	30	0.004900	0.003200	0.001300	0.005600	26	0	4	0.802200	0.199100
augmented_union	support	porto	nsga2 vs nbi_C	hv_ratio	30	0.052400	0.028000	0.005200	0.068300	24	0	6	0.724700	0.250800
augmented_union	support	porto	nsga2 vs nbi_B	igd_plus	30	0.018400	0.011400	0.005000	0.015100	30	0	0	1.000000	0.294100
augmented_union	support	porto	nsga2 vs nbi_B	hv_ratio	30	0.150800	0.114100	0.065300	0.131900	28	0	2	0.969900	0.117300
augmented_union	support	porto	nsga2 vs ws_random_scalarization	igd_plus	30	1.580100	0.092900	0.076100	0.125100	30	0	0	1.000000	0.703600
augmented_union	support	porto	nsga2 vs ws_random_scalarization	hv_ratio	30	0.218200	0.154600	0.103200	0.189100	27	0	3	0.974200	0.147300
augmented_union	support	uci_credit	nsga2 vs nbi_C	igd_plus	30	0.011300	0.013000	0.002800	0.020700	21	0	9	0.660200	0.176800
augmented_union	support	uci_credit	nsga2 vs nbi_C	hv_ratio	30	0.014200	0.017400	0.005500	0.024000	25	0	5	0.724700	0.155300
augmented_union	support	uci_credit	nsga2 vs nbi_B	igd_plus	30	0.041100	0.026500	0.020500	0.033200	26	0	4	0.939800	0.414300
augmented_union	support	uci_credit	nsga2 vs nbi_B	hv_ratio	30	0.085900	0.058200	0.045200	0.081200	28	0	2	0.978500	0.278200
augmented_union	support	uci_credit	nsga2 vs ws_random_scalarization	igd_plus	30	-0.016000	-0.014900	-0.018700	-0.012200	1	0	29	-0.982800	0.003500
augmented_union	support	uci_credit	nsga2 vs ws_random_scalarization	hv_ratio	30	-0.024000	-0.020200	-0.028300	-0.017000	1	0	29	-0.991400	0.003500
augmented_union	weighted	santander	nsga2 vs nbi_C	igd_plus	30	0.007400	0.006000	0.003900	0.007400	29	0	1	0.995700	0.024500
augmented_union	weighted	santander	nsga2 vs nbi_C	hv_ratio	30	0.010700	0.008300	0.006300	0.011000	30	0	0	1.000000	0.012800
augmented_union	weighted	santander	nsga2 vs nbi_B	igd_plus	30	0.008700	0.007900	0.005100	0.010200	30	0	0	1.000000	0.002400
augmented_union	weighted	santander	nsga2 vs nbi_B	hv_ratio	30	0.018100	0.017000	0.015000	0.020100	30	0	0	1.000000	0.000000
augmented_union	weighted	santander	nsga2 vs ws_random_scalarization	igd_plus	30	0.005500	0.005500	0.004300	0.006100	30	0	0	1.000000	0.000100
augmented_union	weighted	santander	nsga2 vs ws_random_scalarization	hv_ratio	30	0.020500	0.019300	0.017100	0.021800	30	0	0	1.000000	0.000000
augmented_union	weighted	bnp	nsga2 vs nbi_C	igd_plus	30	0.008900	0.008700	0.007800	0.009400	30	0	0	1.000000	0.000000
augmented_union	weighted	bnp	nsga2 vs nbi_C	hv_ratio	30	0.013100	0.012700	0.011300	0.013900	30	0	0	1.000000	0.000000
augmented_union	weighted	bnp	nsga2 vs nbi_B	igd_plus	30	0.007600	0.007900	0.006700	0.009000	28	1	1	0.974200	0.014200
augmented_union	weighted	bnp	nsga2 vs nbi_B	hv_ratio	30	0.021900	0.024600	0.020800	0.026500	29	0	1	0.995700	0.003900
augmented_union	weighted	bnp	nsga2 vs ws_random_scalarization	igd_plus	30	0.388700	0.291000	0.277300	0.300900	30	0	0	1.000000	0.035500
augmented_union	weighted	bnp	nsga2 vs ws_random_scalarization	hv_ratio	30	0.551400	0.522300	0.505800	0.538300	30	0	0	1.000000	0.000000
augmented_union	weighted	porto	nsga2 vs nbi_C	igd_plus	30	0.009000	0.004800	0.003500	0.007500	26	0	4	0.944100	0.188300
augmented_union	weighted	porto	nsga2 vs nbi_C	hv_ratio	30	0.020900	0.011500	0.007000	0.016900	26	0	4	0.939800	0.183100
augmented_union	weighted	porto	nsga2 vs nbi_B	igd_plus	30	0.050400	0.034000	0.020500	0.043500	30	0	0	1.000000	0.120100
augmented_union	weighted	porto	nsga2 vs nbi_B	hv_ratio	30	0.125000	0.078400	0.052300	0.110400	30	0	0	1.000000	0.115500
augmented_union	weighted	porto	nsga2 vs ws_random_scalarization	igd_plus	30	1.550000	0.098900	0.069500	0.118500	30	0	0	1.000000	0.708000
augmented_union	weighted	porto	nsga2 vs ws_random_scalarization	hv_ratio	30	0.290100	0.232000	0.174700	0.279800	30	0	0	1.000000	0.041100
augmented_union	weighted	uci_credit	nsga2 vs nbi_C	igd_plus	30	0.003400	0.003200	0.001100	0.004500	23	2	5	0.866700	0.065600
augmented_union	weighted	uci_credit	nsga2 vs nbi_C	hv_ratio	30	0.012100	0.011400	0.007100	0.015600	30	0	0	1.000000	0.005700
augmented_union	weighted	uci_credit	nsga2 vs nbi_B	igd_plus	30	0.159200	0.143200	0.128200	0.183600	30	0	0	1.000000	0.002100
augmented_union	weighted	uci_credit	nsga2 vs nbi_B	hv_ratio	30	0.287500	0.279200	0.228100	0.334700	30	0	0	1.000000	0.001300
augmented_union	weighted	uci_credit	nsga2 vs ws_random_scalarization	igd_plus	30	0.004400	0.002600	0.001900	0.004200	27	2	1	0.978500	0.120900
augmented_union	weighted	uci_credit	nsga2 vs ws_random_scalarization	hv_ratio	30	0.013400	0.009600	0.007500	0.012500	30	0	0	1.000000	0.043200
sample_core	support	santander	nsga2 vs nbi_C	igd_plus	30	-0.000100	-0.000300	-0.000900	0.000900	12	0	18	0.002200	0.879500
sample_core	support	santander	nsga2 vs nbi_C	hv_ratio	30	0.003300	0.002400	0.000800	0.004200	29	0	1	0.883900	0.179900
sample_core	support	santander	nsga2 vs nbi_B	igd_plus	30	0.005700	0.005800	0.005100	0.006300	30	0	0	1.000000	0.000000

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reference	cost	dataset	comparison	endpoint	n	mean	median	ci95 median lo	ci95 median hi	wins	ties	losses	rank biserial	p nb
sample_core	support	santander	nsga2 vs nbi_B	hv_ratio	30	0.009000	0.011200	0.004900	0.012300	29	0	1	0.978500	0.007600
sample_core	support	santander	nsga2 vs ws_random_scalarization	igd_plus	30	0.007600	0.007400	0.006600	0.008000	30	0	0	1.000000	0.000800
sample_core	support	santander	nsga2 vs ws_random_scalarization	hv_ratio	30	0.011900	0.011100	0.010500	0.012100	29	0	1	0.995700	0.000400
sample_core	support	bnp	nsga2 vs nbi_C	igd_plus	30	0.001400	0.000800	0.000300	0.001300	25	0	5	0.836600	0.213100
sample_core	support	bnp	nsga2 vs nbi_C	hv_ratio	30	0.015000	0.017800	0.016800	0.018700	30	0	0	1.000000	0.000100
sample_core	support	bnp	nsga2 vs nbi_B	igd_plus	30	0.004700	0.004900	0.003100	0.006400	24	0	6	0.858100	0.059800
sample_core	support	bnp	nsga2 vs nbi_B	hv_ratio	30	0.009000	0.010200	0.004300	0.014900	25	0	5	0.896800	0.023300
sample_core	support	bnp	nsga2 vs ws_random_scalarization	igd_plus	30	0.468500	0.371000	0.349300	0.382100	30	0	0	1.000000	0.019100
sample_core	support	bnp	nsga2 vs ws_random_scalarization	hv_ratio	30	0.522900	0.483200	0.467700	0.503100	30	0	0	1.000000	0.000000
sample_core	support	porto	nsga2 vs nbi_C	igd_plus	30	0.008100	0.004800	0.002000	0.007500	25	1	4	0.793500	0.228300
sample_core	support	porto	nsga2 vs nbi_C	hv_ratio	30	0.052800	0.028200	0.005200	0.068800	24	0	6	0.724700	0.249400
sample_core	support	porto	nsga2 vs nbi_B	igd_plus	30	0.027900	0.013800	0.007200	0.020100	29	0	1	0.982800	0.282300
sample_core	support	porto	nsga2 vs nbi_B	hv_ratio	30	0.151800	0.115500	0.065500	0.131100	28	0	2	0.969900	0.115900
sample_core	support	porto	nsga2 vs ws_random_scalarization	igd_plus	30	1.528800	0.050500	0.041300	0.081300	30	0	0	1.000000	0.711800
sample_core	support	porto	nsga2 vs ws_random_scalarization	hv_ratio	30	0.219000	0.154000	0.103800	0.190700	27	0	3	0.974200	0.146600
sample_core	support	uci_credit	nsga2 vs nbi_C	igd_plus	30	0.009900	0.009500	0.001400	0.020300	21	0	9	0.617200	0.225100
sample_core	support	uci_credit	nsga2 vs nbi_C	hv_ratio	30	0.014100	0.017400	0.005500	0.024000	25	0	5	0.724700	0.156500
sample_core	support	uci_credit	nsga2 vs nbi_B	igd_plus	30	0.042500	0.027400	0.018200	0.030800	27	1	2	0.935500	0.438300
sample_core	support	uci_credit	nsga2 vs nbi_B	hv_ratio	30	0.085800	0.058200	0.045200	0.081200	28	0	2	0.978500	0.278800

Table S23: Median indicator levels for every set under both reference definitions. First 70 of 128 rows; the complete table is the source CSV. Showing 10 of 13 columns; the rest are in the source CSV.

reference	cost	dataset	set	igd plus	hv ratio	gd front	spacing	spacing cv	n front
augmented_union	support	bnp	design_runs	0.037600	0.964300	0.009900	0.524700	1.132000	5.000000
augmented_union	support	bnp	nbi_A	0.042200	0.907600	0.004400	0.264600	5.196200	27.000000
augmented_union	support	bnp	nbi_B	0.029900	0.856900	0.013600	0.473100	2.030800	10.000000
augmented_union	support	bnp	nbi_C	0.037900	0.811500	0.006800	0.509300	2.989300	13.000000
augmented_union	support	bnp	nsga2	0.016200	0.873200	0.003900	0.097200	2.530400	16.000000
augmented_union	support	bnp	random_dirichlet_budget	1.026100	0.023300	6.615500	1.234700	1.177700	2.000000
augmented_union	support	bnp	single_objective_refs	0.048700	0.808500	0.001600	0.542300	1.510400	6.000000
augmented_union	support	bnp	ws_random_scalarization	0.349400	0.532900	0.005700	0.000000	0.738900	66.000000
augmented_union	support	porto	design_runs	0.159900	0.709600	0.088500	0.594900	0.768400	3.000000
augmented_union	support	porto	nbi_A	0.094300	0.837300	0.045900	0.540500	0.952900	3.000000
augmented_union	support	porto	nbi_B	0.016700	0.868900	0.027900	0.455300	1.300300	6.000000
augmented_union	support	porto	nbi_C	0.007300	0.934000	0.012100	0.337100	2.058900	12.500000
augmented_union	support	porto	nsga2	0.003800	0.987900	0.003800	0.329900	2.128900	12.000000
augmented_union	support	porto	random_dirichlet_budget	1.230300	0.000900	6.644700	47.465700	1.538700	2.000000
augmented_union	support	porto	single_objective_refs	0.013300	0.857600	0.008800	0.528300	2.163900	5.000000
augmented_union	support	porto	ws_random_scalarization	0.095400	0.831800	0.056400	0.431600	1.999300	4.500000
augmented_union	support	santander	design_runs	0.008000	0.987800	0.003800	0.847200	1.664800	4.000000
augmented_union	support	santander	nbi_A	0.066700	0.852900	0.078900	0.444700	1.838200	8.000000
augmented_union	support	santander	nbi_B	0.008700	0.985800	0.021600	0.440800	2.011200	11.000000
augmented_union	support	santander	nbi_C	0.001600	0.994600	0.018100	0.256500	3.166900	29.500000
augmented_union	support	santander	nsga2	0.002000	0.997600	0.003900	0.094600	2.957300	22.000000
augmented_union	support	santander	random_dirichlet_budget	0.648200	0.076600	0.251200	0.465400	1.105300	4.000000
augmented_union	support	santander	single_objective_refs	0.011800	0.984100	0.001500	0.776000	0.924900	4.000000
augmented_union	support	santander	ws_random_scalarization	0.009900	0.986300	0.003900	0.019100	1.802400	11.500000
augmented_union	support	uci_credit	design_runs	0.012600	0.975000	0.002300	0.221400	0.338700	4.000000
augmented_union	support	uci_credit	nbi_A	0.039000	0.910700	0.027500	0.395500	2.048800	8.500000
augmented_union	support	uci_credit	nbi_B	0.047800	0.895400	0.024200	0.290000	2.262600	14.000000
augmented_union	support	uci_credit	nbi_C	0.036000	0.941600	0.007900	0.373700	2.770100	16.500000
augmented_union	support	uci_credit	nsga2	0.020700	0.959500	0.029600	0.214700	1.731000	12.000000
augmented_union	support	uci_credit	random_dirichlet_budget	0.557200	0.097300	0.131300	1.149100	1.501800	2.000000
augmented_union	support	uci_credit	single_objective_refs	0.036700	0.905400	0.001600	0.866200	2.285800	6.000000
augmented_union	support	uci_credit	ws_random_scalarization	0.005500	0.983100	0.001300	0.096700	2.869400	15.000000
augmented_union	weighted	bnp	design_runs	0.031100	0.917200	0.091700	0.522500	1.039800	6.000000
augmented_union	weighted	bnp	nbi_A	0.027400	0.913500	0.001600	0.207600	0.825300	8.000000
augmented_union	weighted	bnp	nbi_B	0.012400	0.969200	0.001200	0.130900	2.967000	44.000000
augmented_union	weighted	bnp	nbi_C	0.012800	0.981500	0.008600	0.076500	0.882800	34.000000
augmented_union	weighted	bnp	nsga2	0.004200	0.993900	0.000800	0.019900	0.618500	66.000000
augmented_union	weighted	bnp	random_dirichlet_budget	1.645800	0.000000	5.294500	3.627800	0.981500	5.000000
augmented_union	weighted	bnp	single_objective_refs	0.154400	0.752100	0.025600	0.441900	0.486900	4.000000
augmented_union	weighted	bnp	ws_random_scalarization	0.295100	0.471900	0.084000	0.036200	1.575400	1.000000
augmented_union	weighted	porto	design_runs	0.148200	0.645400	0.063900	0.597800	0.779100	3.000000
augmented_union	weighted	porto	nbi_A	0.095300	0.766400	0.035800	0.067600	0.497900	11.000000
augmented_union	weighted	porto	nbi_B	0.035900	0.915200	0.014600	0.062400	2.575000	59.500000
augmented_union	weighted	porto	nbi_C	0.007700	0.980700	0.004100	0.047300	1.556400	56.000000
augmented_union	weighted	porto	nsga2	0.002100	0.995100	0.001300	0.020000	0.602600	66.000000
augmented_union	weighted	porto	random_dirichlet_budget	1.683700	0.000000	46.478200	32.401200	1.113500	5.000000
augmented_union	weighted	porto	single_objective_refs	0.082500	0.823100	0.007700	0.385200	1.442300	5.000000
augmented_union	weighted	porto	ws_random_scalarization	0.102600	0.760800	0.034700	0.107200	2.160200	12.500000
augmented_union	weighted	santander	design_runs	0.009600	0.980500	0.001900	0.308900	1.006100	7.000000
augmented_union	weighted	santander	nbi_A	0.119100	0.802900	0.611600	0.978200	1.637900	13.000000
augmented_union	weighted	santander	nbi_B	0.010200	0.981000	0.014600	0.153100	1.064000	19.500000

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reference	cost	dataset	set	igd plus	hv ratio	gd front	spacing	spacing cv	n front
augmented_union	weighted	santander	nbi_C	0.008700	0.989700	0.009100	0.198700	2.283400	35.000000
augmented_union	weighted	santander	nsga2	0.003200	0.997800	0.003200	0.026000	0.737300	66.000000
augmented_union	weighted	santander	random_dirichlet_budget	0.634500	0.268400	4.253200	1.598000	0.915000	6.000000
augmented_union	weighted	santander	single_objective_refs	0.014100	0.970300	0.020700	0.752300	0.899200	4.000000
augmented_union	weighted	santander	ws_random_scalarization	0.008000	0.978400	0.000700	0.015600	2.846000	57.000000
augmented_union	weighted	uci_credit	design_runs	0.032100	0.921400	0.012600	0.299500	1.100100	8.000000
augmented_union	weighted	uci_credit	nbi_A	0.210300	0.633100	0.006900	0.123000	1.402900	11.000000
augmented_union	weighted	uci_credit	nbi_B	0.150000	0.705900	0.002400	0.063200	1.950400	27.000000
augmented_union	weighted	uci_credit	nbi_C	0.009300	0.976600	0.001400	0.031300	1.919900	65.000000
augmented_union	weighted	uci_credit	nsga2	0.006300	0.988300	0.004800	0.028300	0.649800	66.000000
augmented_union	weighted	uci_credit	random_dirichlet_budget	0.149100	0.729300	0.285100	0.375700	0.934200	9.500000
augmented_union	weighted	uci_credit	single_objective_refs	0.401300	0.336900	0.022100	1.134700	1.830600	5.000000
augmented_union	weighted	uci_credit	ws_random_scalarization	0.008700	0.978100	0.000500	0.029300	1.511900	43.500000
sample_core	support	bnp	design_runs	0.040400	0.932600	0.010700	0.546800	1.303300	5.000000
sample_core	support	bnp	nbi_A	0.045200	0.895100	0.007400	0.264600	5.196200	27.000000
sample_core	support	bnp	nbi_B	0.016100	0.978300	0.012500	0.460300	2.140200	10.000000
sample_core	support	bnp	nbi_C	0.011500	0.970300	0.009100	0.491500	3.034800	13.000000
sample_core	support	bnp	nsga2	0.010600	0.988400	0.006000	0.036000	2.116400	16.000000
sample_core	support	bnp	random_dirichlet_budget	0.891100	0.072400	0.424000	0.803800	1.006700	2.000000

Table S24: NSGA-II realized evaluation and wall-clock ratios against NBI-C. The evaluation columns are in different units: NSGA-II counts weight-vector evaluations and NBI-C counts each ROC-AUC and each log-loss computation (two per weight-vector evaluation), so a ratio of 1 means NSGA-II performed twice as many weight-vector evaluations; wall clock is not matched either. Showing 9 of 11 columns; the rest are in the source CSV.

dataset	n	nsga2 evals total	nbi c evals total	eval ratio median	nsga2 seconds mean	nbi c seconds mean	time ratio median	n nondominated median
bnp	30	12508320	12508244	1.000000	3040.359300	1408.892900	2.156800	66.000000
porto	30	12505284	12505392	1.000000	5541.422900	2502.923300	2.211900	66.000000
santander	30	12361074	12361066	1.000000	5526.433600	2526.775700	2.184100	66.000000
uci_credit	30	12775488	12775674	1.000000	689.333500	306.357400	2.251700	66.000000

The pre-specified configuration is `nsga2_preregistered_config.json`, committed before the first run. The full report, including the ten verification checks, is `reports/pco213_postwork_benchmark/nsga2/NSGA2_BASELINE_REPORT.md`. That report describes the budget as matched to NBI-C's evaluations; the caption of Table S24 states why the two evaluation counts are in different units, and a dated erratum filed beside the report (`NSGA2_BASELINE_ERRATUM.md`) records the correction.

S14 Corrections log

Readings that appeared in earlier analyses of this study and did not survive replication or adversarial verification, with the evidence that overturned each. They are listed because each is traceable in the repository history and a reader should be able to see what changed.

1. **The BNP Paribas support-cost bimodality of NBI-C** was reported at $R = 10$ as a regime. It is a per-replication normalization-box artifact (Section S9).

2. **A Santander deployment-cost gap between the AUC and log-loss optima** was attributed to the AUC optimum. It is an activity-threshold artifact: the derivative-free search leaves a k-nearest-neighbour weight just above 10^{-3} that is AUC-inert, and an AUC-equivalent blend without it costs about +4 ms per 1,000 rows.

3. **The random Dirichlet comparator was described as budget-matched.** It is matched only in the number of returned candidates and is degenerate under the weighted cost; it is a floor check.

4. **Uniform spacing was treated as an advantage of NBI.** It does not survive revalidation, and the NSGA-II arm shows the shortfall is a property of the CHIM construction rather than of the objective geometry.

5. **The gate-conditional anchor gain on BNP Paribas** was read as evidence that real anchors add more when the ROC-AUC surface is reliable. It restates NBI-A's collapse, since three of the five partitions that pass the ROC-AUC gate are collapse partitions.

6. **The A-versus-B contrast was described as a clean single-factor experiment.** It is not: replacing the anchors also moves the utopia point, rescales both normalization axes, reorients the CHIM and substitutes three returned candidates (Section S12).

7. **The compute premium was quoted from solver stages only.** Charged standalone, so that each arm carries the stages it depends on, the NBI-C premium over NBI-B is 2.3–6.2×, not 3–73×.

8. **The NSGA-II baseline was described as matched to NBI-C in real objective evaluations.** Its budget was set to NBI-C's count of real objective evaluations, which records each ROC-AUC and each log-loss computation separately, two per weight-vector evaluation, whereas NSGA-II counts weight-vector evaluations; NSGA-II therefore performed twice as many weight-vector evaluations as NBI-C's solve, and its 2.2× wall clock is close to that ratio rather than a consequence of batching (Section S13).